



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION (Regulation R19) **M.TECH (COMPUTER SCIENCE & TECHNOLOGY)** **DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING** (With effect from 2019-2020 Admitted Batch onwards) **I-SEMESTER**

Subject Code.	Name of the Subject	Category.	C	L	T	P	Internal Marks	External Marks	Total Marks
M19 CST 1101	Mathematical Foundations of Computer Science	PC	3	3	0	0	25	75	100
M19 CST 1102	Advanced Data Structures	PC	3	3	0	0	25	75	100
#PE-I	Program Elective-I	PE	3	3	0	0	25	75	100
#PE-II	Program Elective-II	PE	3	3	0	0	25	75	100
M19 RD 1101	Research Methodology and IPR	CC	2	2	0	0	25	75	100
M19 CST 1113	Advanced Data Structures Lab	PC	2	0	0	4	25	75	100
M19 CST 1114	Computing Lab	PC	2	0	0	4	25	75	100
#AC-1	Audit course -1	AC	0	2	0	0	0	0	0
Total			18	16	0	8	175	525	700

	Code	Course		Code	Course
#PE-I	M19CST1103	Artificial Intelligence	# AC 1 & 2	M19AC0001	English for Research Paper Writing
	M19CST1104	Cloud Computing		M19AC0002	Disaster Management
	M19CST1105	Digital Image Processing		M19AC0003	Sanskrit for Technical Knowledge
	M19CST1106	Advanced Operating System		M19AC0004	Value Education
	M19CST1107	Optimization Techniques		M19AC0005	Constitution of India
#PE-II	M19CST1108	Bigdata Analytics		M19AC0006	Pedagogy Studies
	M19CST1109	Applied Cryptography		M19AC0007	Stress Management by Yoga
	M19CST1110	Advanced Computer Networks		M19AC0008	Personality Development through Life Enlightenment Skills.
	M19CST1111	Embedded Computing			
	M19CST1112	Parallel Computer Architecture			

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1101	PC	3	0	0	3	25	75	3 Hrs.
MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE								
Course Objectives:								
1.	To understand the mathematical fundamentals that is prerequisites for a variety of courses like Data mining, Network protocols, analysis of Web traffic, Computer security, Software engineering, Computer architecture, operating systems, distributed systems, Bioinformatics, Machine learning.							
2.	To develop the understanding of mathematical and logical basis to many modern techniques in information technology like machine learning, programming language design and concurrency.							
3.	To study various sampling and classification problems.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	To understand the basic notions of discrete and continuous probability.							K2
2.	To understand the methods of statistical inference, and the role that sampling distributions play in those methods.							K2
3.	To be able to perform correct and meaningful statistical analyses of simple to moderate complexity.							K3
4.	Fit a best suitable curve for the given data							K3
5.	Utilize the concepts in graph theory in their field.							K3
6.	Solve different counting problems as well as recurrence relations.							K2
SYLLABUS								
UNIT-I (12 Hrs)	Review on basic Probability concepts- sample space -Addition and Multiplication theorems- Bay’s theorem- Random variables (Discrete & Continuous). Probability mass, density and cumulative distribution functions(for discrete & continuous random variables)-Joint distribution function- discrete & continuous random variables-conditional distribution- Expected value, Variance- Conditional expectation- Central limit theorem, probabilistic inequalities- Chebychev’s inequality, Jenson’s inequality- Cauchy – Schwartz inequatlity- Convergence in probability- Markov chains.							
UNIT-II (10 Hrs)	Parametric families of distributions: Binomial, Poisson, Normal, Uniform, exponential and Gamma distribution, derivation of basic characteristics, Hypergeometric& multinomial distributions.							
UNIT-III (10 Hrs)	Statistical inference, introduction to multivariate Statistical models: Scatter diagram, fitting of straight line, parabola, power and exponential curves: ax^b, ab^x, ae^{bx} using method of least square approximation, multiple regression (linear equations)- principal component analysis. Random Samples: Sampling distributions of Estimators, efficient and unbiased estimation, point & interval estimator: Method of moments and maximum likelyhood estimator.							

UNIT-IV (10 Hrs)	Graph Theory: Isomorphism, Planar graphs, Euler's and Hamiltonian circuits or Euler's cycles: graph coloring- Chromatic number. Trees: Introduction- Trees- Labeled tree- some diagrams on directed and undirected trees, Review of Basic properties of a tree- Sequential representation of a binary tree.
UNIT-V (12 Hrs)	Counting Methods: Permutation & Combination with & without repetition: Principal inclusion & exclusion: Special technique to solve combinational enumerator problem: Recurrence relation: order & degree of recurrence relation, Solution of Linear homogeneous and non-homogeneous recurrence relation with constant coefficient using methods of generating function, characteristic roots & particular solution by method of determinant coefficients.
Text Books:	
1.	Murray R. Spiegel, Probability & Statistics, Schaum's Outline Series McGraw-Hill.
2.	Rosen K.H., Discrete Mathematical Structure, Sixth Edition, McGraw-Hill- International Edition.
3.	Miller and Freund's, Probability & Statistics for Engineers, Eighth Edition, By Pearson.
4.	Swapna Kumar chakraborty and Bikashkantisarkar, Discrete Mathematics, Oxford Higher Education.
Reference Books:	
1.	Alan Tucker, Applied Combinatorics, 6 th Edition, Wiley.
2.	C.L.Liu, Elements of Discrete Mathematics, 2 nd Edition, McGraw-Hill Book Company
3.	John Vince, Foundation Mathematics for Computer Science, Springer
4.	K. Trivedi, Probability and Statistics with reliability, Queuing and Computer Science Applications, 2 nd edition, John Wiley and Sons Ltd. Chichester, UK.
5.	M. Mitzenmacher and E. Upfal, Probability and Computing: Randomized Algorithms and Probabilistic Analysis, Cambridge University Press.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1102	PC	3	0	0	3	25	75	3 Hrs.
ADVANCED DATA STRUCTURES								
Course Objectives:								
1.	The student should be able to choose appropriate data structures, understand the ADT/libraries, and use it to design algorithms for a specific problem.							
2.	Students should be able to understand the necessary mathematical abstraction to solve problems.							
3.	To familiarize students with advanced paradigms and data structure used to solve algorithmic problems.							
4.	Student should be able to come up with analysis of efficiency and proofs of correctness.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand the implementation of symbol table using hashing techniques.							K2
2.	Understand the implementation of skip lists and Need for Randomizing Data Structure							K2
3.	Develop and analyze algorithms for red-black trees, B-trees and Splay trees.							K3
4.	Develop algorithms for text processing applications.							K3
5.	Identify suitable data structures and develop algorithms for computational geometry problems.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Dictionaries: Definition, Dictionary Abstract Data Type, Implementation of Dictionaries. Hashing:Review of Hashing, Hash Function, Collision Resolution Techniques in Hashing, Separate Chaining, Open Addressing, Linear Probing, Quadratic Probing, Double Hashing, Rehashing, Extendible Hashing.							
UNIT-II (10 Hrs)	Skip Lists: Need for Randomizing Data Structures and Algorithms, Search and Update Operations on Skip Lists, Probabilistic Analysis of Skip Lists, Deterministic Skip Lists							
UNIT-III (10Hrs)	Trees: Binary Search Trees, AVL Trees, Red Black Trees, 2-3 Trees, B-Trees, Splay Trees							
UNIT-IV (10 Hrs)	Text Processing: Sting Operations, Brute-Force Pattern Matching, The Boyer-Moore Algorithm, The Knuth-Morris-Pratt Algorithm, Standard Tries, Compressed Tries, Suffix Tries, The Huffman Coding Algorithm, The Longest Common Subsequence Problem (LCS), Applying Dynamic Programming to the LCS Problem.							
UNIT-V (10Hrs)	Computational Geometry: One Dimensional Range Searching, Two Dimensional Range Searching, Constructing a Priority Search Tree, Searching a Priority Search Tree, Priority Range Trees, Quad trees, k- D Trees. Recent Trends in Hashing, Trees, and various computational geometry methods for efficiently solving the new evolving problem stochastic process in manufacturing, Game theory							

Text Books:	
1.	J Mark Allen Weiss, Data Structures and Algorithm Analysis in C++, 2nd Edition, Pearson, 2004
2.	M T Goodrich, Roberto Tamassia, Algorithm Design, John Wiley, 2002.
Reference Books:	
1.	“Advanced Data Structures” by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1103	PE	3	0	0	3	25	75	3 Hrs.
ARTIFICIAL INTELLIGENCE								
Course Objectives:								
1.	Gain a historical perspective of AI and its foundations.							
2.	Become familiar with basic principles of AI toward problem solving, inference, perception knowledge representation, and learning.							
3.	Investigate applications of AI techniques in intelligent agents, expert systems, artificial neural networks and other machine learning models.							
4.	Experience AI development tools such as an „AI language“, expert system shell, and/or data mining tool. Experiment with a machine learning model for simulation and analysis.							
5.	Explore the current scope, potential, limitations, and implications of intelligent systems.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Demonstrate knowledge of the building blocks of AI as presented in terms of intelligent agents							K2
2.	Analyze and formalize the problem as a state space, graph, design heuristics and select amongst different search or game based techniques to solve them.							K2
3.	Develop intelligent algorithms for constraint satisfaction problems and also design intelligent systems for Game Playing							K3
4.	Attain the capability to represent various real life problem domains using logic based techniques and use this to perform inference or planning.							K3
5.	Solve problems with uncertain information using Bayesian approaches.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to artificial intelligence: Introduction , history, intelligent systems, foundations of AI, applications, tic-tac-tie game playing, development of AI languages, current trends in AI, Problem solving: state-space search and control strategies: Introduction, general problem solving, characteristics of problem, exhaustive searches, heuristic search techniques, iterative- deepening a*, constraint satisfaction							
UNIT-II (10 Hrs)	Problem reduction and game playing: Introduction, problem reduction, game playing, alpha- beta pruning, two-player perfect information games, Logic concepts: Introduction, propositional calculus, propositional logic, natural deduction system, axiomatic system, semantic tableau system in propositional logic, resolution refutation in propositional logic, predicate logic							
UNIT-III (10Hrs)	Knowledge representation: Introduction, approaches to knowledge representation, knowledge representation using semantic network, extended semantic networks for KR, knowledge representation using frames, advanced knowledge representation techniques:							

	Introduction, conceptual dependency theory, script structure, cyc theory, case grammars, semantic web, Expert system and applications: Introduction phases in building expert systems, expert system versus traditional systems, rule-based expert systems blackboard systems truth maintenance systems, application of expert systems, list of shells and tools
UNIT-IV (10 Hrs)	Uncertainty measure: probability theory: Introduction, probability theory, Bayesian belief networks, certainty factor theory, dempster-shafer theory , Fuzzy sets and fuzzy logic: Introduction, fuzzy sets, fuzzy set operations, types of membership functions, multi valued logic, fuzzy logic, linguistic variables and hedges, fuzzy propositions, inference rules for fuzzy propositions, fuzzy systems.
UNIT-V (10Hrs)	Machine learning paradigms: Introduction, machine learning systems, supervised and unsupervised learnings, inductive learning, deductive learning, clustering, support vector machines, case based reasoning and learning, Artificial neural networks: Introduction, artificial networks, single layer feed forward networks, multi layered forward networks, design issues of artificial neural networks
Text Books:	
1.	Artificial Intelligence- SarojKaushik, CENGAGELearning,
2.	Artificial intelligence, A modern Approach, 2nd ed, Stuart Russel, Peter Norvig, PEA
3.	Artificial Intelligence- 3rd ed, Rich, Kevin Knight, Shiv Shankar B Nair, TMH
4.	Introduction to Artificial Intelligence, Patterson, PHI
Reference Books:	
1.	Artificial intelligence, structures and Strategies for Complex problem solving, 5th ed, George F Lugar,PEA
2.	Introduction to Artificial Intelligence, Ertel, Wolf Gang, Springer
3.	Artificial Intelligence, A new Synthesis, Nils J Nilsson, Elsevier

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1104	PE	3	0	0	3	25	75	3 Hrs.
CLOUD COMPUTING								
Course Objectives:								
1.	To provide an in-depth and comprehensive knowledge of the Cloud Computing fundamental issues, technologies, applications and implementations.							
2.	To expose the students to the frontier areas of Cloud Computing							
3.	To motivate students to do programming and experiment with the various cloud computing environments							
4.	To shed light on the Security issues in Cloud Computing							
5.	To introduce about the Cloud Standards							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Articulate the main concepts, key technologies, strengths, and limitations of cloud computing and the possible applications for state-of-the-art cloud computing.							K1
2.	Use and Examine different cloud computing services							K3
3.	Identify the appropriate cloud platform and software environment for the given application.							K3
4.	Explain the core issues of cloud computing such as security, privacy, and interoperability.							K2
SYLLABUS								
UNIT-I (8 Hrs)		History of Centralized and Distributed Computing, Overview of Distributed Computing, Cluster computing, Grid computing. Technologies for Network based systems, System models for Distributed and cloud computing, Software environments for distributed systems and clouds.						
UNIT-II (8 Hrs)		Introduction to Cloud Computing, Cloud issues and challenges, Properties, Characteristics, Service models, Deployment models. Cloud resources: Network and API, Virtual and Physical computational resources, Data-storage. Virtualization concepts, Types of Virtualization: Introduction to Various Hypervisors, High Availability (HA)/Disaster Recovery (DR) using Virtualization, Moving VMs.						
UNIT-III (12 Hrs)		Service models, Infrastructure as a Service (IaaS), Resource Virtualization: Server, Storage, Network, Case studies. Platform as a Service (PaaS), Cloud platform & Management: Computation, Storage, Case studies. Software as a Service (SaaS), Web services, Web 2.0, Web OS, Case studies, Anything as a service (XaaS).						
UNIT-IV (12 Hrs)		Cloud Programming and Software Environments, Parallel and Distributed Programming paradigms, Programming on Amazon AWS and Microsoft Azure, Programming support of Google App Engine, Emerging Cloud software Environment						

UNIT-V (8 Hrs)	Cloud Access: authentication, authorization and accounting, Cloud Provenance and meta-data, Cloud Reliability and fault-tolerance, Cloud Security, privacy, policy and compliance, Cloud federation, interoperability and standards.
Text Book:	
1.	Kai Hwang, Geoffrey C. Fox and Jack J. Dongarra, “Distributed and cloud computing from Parallel Processing to the Internet of Things”, Morgan Kaufmann, Elsevier – 2012
Reference Books:	
1.	Barrie Sosinsky, “Cloud Computing Bible” John Wiley & Sons, 2010
2.	Tim Mather, SubraKumaraswamy, and ShahedLatif, Cloud Security and Privacy An Enterprise Perspective on Risks and Compliance, O'Reilly 2009

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1105	PE	3	0	0	3	25	75	3 Hrs.
DIGITAL IMAGE PROCESSING								
(Program Elective-1)								
Course Objectives:								
1.	To understand basic algorithms for image processing.							
2.	To understand various filters, Point processing, and Arithmetic operations in image processing.							
3.	To introduce key concepts in pattern recognition and machine learning.							
4.	To understand different applications of graphics and image processing.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Students are able to develop software tools such as Games, Animation, and Recognition system.							K3
2.	Key concepts, tools and approaches for pattern recognition on complex datasets.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Digital Image Fundamentals: Image Formation and types, Basic geometric transformations, Fourier Transforms, Walsh, Hadamard, Discrete Cosine, Hotelling Transforms.							
UNIT-II (10 Hrs)	Image Enhancement and Restoration: Histogram Modification Techniques, Image Smoothing, Image Sharpening, Image Restoration, Degradation Model, Noise Models, Spatial Filtering, Frequency Domain Filtering.							
UNIT-III (10 Hrs)	Image Segmentation and Recognition: Detection of Discontinuities, Edge Linking and Boundary Detection, Thresholding, Region Based Segmentation, Morphology operations. Pattern classification, Clustering and Matching, Knowledge representation and use for scene analysis and image understanding (2D and 3D),Object recognition and identification, Case study of various applications							
UNIT-IV (10 Hrs)	Pattern Recognition: Key concepts, Supervised/Unsupervised Learning, Loss functions and generalization, Probability Theory, Parametric VS Non-parametric methods, Elements of Computational Learning Theory Ensemble Learning, Bagging, Boosting, Random Forest							
UNIT-V (8 Hrs)	Dimensionality Reduction - CCA, LDA, ICA, NMF - Canonical Variates - Feature Selection vs Feature Extraction, Filter Methods - Sub-space approaches - Embedded methods, Low-Rank approaches - Recommender Systems							
Text Books:								
1.	Rafael C. Gonzalez, Richard E. Woods, “Digital Image Processing”, Pearson Education, Third edition,2011.							
2.	Christopher M. Bishop, Pattern Recognition and MachineLearning.							
3.	John Shawe-Taylor and NelloCristianini, Kernel Methods for Pattern.							
Reference Books:								
1.	Anil Jain K, “Fundamentals of Digital Image Processing”, Prentice-Hall of India, 2001.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1106	PE	3	0	0	3	25	75	3 Hrs.
ADVANCED OPERATING SYSTEMS								
(Program Elective-1)								
Course Objectives:								
1.	To study the characteristics of OS for Multiprocessor andMulticomputer.							
2.	To learn the issues related to designingOS.							
3.	To learn the latest trends in building MobileOS.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Knowledge about advanced concepts in OS							K1
2.	Ability to develop OS for distributedsystems							K2
3.	Ability to develop modules for mobiledevices							K2
SYLLABUS								
UNIT-I (8 Hrs)	Multiprocessor Operating Systems: System Architectures, Structures of OS, OS design issues, Process synchronization, Process Scheduling and Allocation, memory management.							
UNIT-II (10 Hrs)	Distributed Operating Systems: System Architectures, Design issues, Communication models, clock synchronization, mutual exclusion, election algorithms, Distributed Deadlock detection							
UNIT-III (8 Hrs)	Distributed scheduling: Distributed shared memory, Distributed File system, Multimedia file systems, File placement, Caching							
UNIT-IV (10 Hrs)	Database Operating Systems: Requirements of Database OS, Transaction process model, Synchronization primitives, Concurrency control algorithms							
UNIT-V (10 Hrs)	Mobile Operating Systems: ARM and Intel architectures, Power Management, Mobile OS Architectures, Underlying OS, Kernel structure and native level programming, Runtime issues, Approaches to power management							
Text Books:								
1.	M Singhal and NG Shivaratri , Advanced Concepts in Operating Systems, Tata McGraw Hill Inc, 2001							
Reference Books:								
1.	A S Tanenbaum, Distributed Operating Systems, Pearson Education Asia,2001							
2.	Source Wikipedia, Mobile Operating Systems, General Books LLC,2010							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1107	PE	3	0	0	3	25	75	3 Hrs.
OPTIMIZATION TECHNIQUES								
(Program Elective-1)								
Course Objectives:								
1.	To understand the theory of optimization methods and algorithms developed for solving various types of optimization problems.							
2.	To develop and promote research interest in applying optimization techniques in problems of Engineering and Technology.							
3.	To apply the mathematical results and numerical techniques of optimization theory to concrete Engineering problems							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Students should able to apply the dynamic programming to solve problems of discreet and continuous variables.							2
2.	Students should able to apply the concept of non-linear programming							2
3.	Students should able to carry out sensitivity analysis							3
4.	Student should able to model the real world problem and simulate it.							3
SYLLABUS								
UNIT-I (8 Hrs)	Optimization Techniques, Model Formulation, models, General L.R Formulation, Simplex Techniques, Sensitivity Analysis, Inventory Control Models							
UNIT-II (10 Hrs)	Formulation of a LPP - Graphical solution revised simplex method - duality theory - dual simplex method - sensitivity analysis - parametric programming							
UNIT-III (8 Hrs)	Nonlinear programming problem - Kuhn-Tucker conditions min cost flow problem - max flow problem - CPM/PERT							
UNIT-IV (10 Hrs)	Scheduling and sequencing - single server and multiple server models - deterministic inventory models - Probabilistic inventory control models - Geometric Programming.							
UNIT-V (10 Hrs)	Competitive Models, Single and Multi-channel Problems, Sequencing Models, Dynamic Programming, Flow in Networks, Elementary Graph Theory, Game Theory Simulation							
Text Books:								
1.	H.A. Taha, Operations Research, An Introduction, PHI,2008							
2.	H.M. Wagner, Principles of Operations Research, PHI, Delhi,1982.							
Reference Books:								
1.	J.C. Pant, Introduction to Optimisation: Operations Research, Jain Brothers, Delhi,2008							
2.	Hitler Libermann Operations Research: McGraw Hill Pub.2009							
3.	Pannerselvam, Operations Research: Prentice Hall of India2010							
4.	Harvey M Wagner, Principles of Operations Research: Prentice Hall of India2010							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1108	PE	3	0	0	3	25	75	3 Hrs.
BIG DATA ANALYTICS								
(Program Elective-2)								
Course Objectives:								
1.	To Provide an overview of an exciting growing field of Big Data analytics.							
2.	To Introduce the tools required to manage and analyze big data like Hadoop, MapReduce etc.,							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand the programming requirements viz., generic types and methods to perform data analysis							K2
2.	Understand the existing technologies and the need of distributed files systems to analyze the big data							K2
3.	To understand and analyze Map-Reduce programming model for better optimization							K2
4.	Collect, manage, store, query, and analyze big data; and identify the need of interfaces to perform I/O operations in Hadoop							K3
5.	Identify the need based tools, viz., Pig and Hive and to handle							K3
6.	Formulate an effective strategy to implement a successful Data analytics project							K3
SYLLABUS								
UNIT-I (10 Hrs)	Data structures in Java: Linked List, Stacks, Queues, Sets, Maps; Generics: Generic classes and Type parameters, Implementing Generic Types, Generic Methods, Wrapper Classes, Concept of Serialization							
UNIT-II (10 Hrs)	Working with Big Data: Google File System, Hadoop Distributed File System (HDFS), Building blocks of Hadoop (Name node, Data node, Secondary Name node, Job Tracker, Task Tracker), Introducing and Configuring Hadoop cluster (Local, Pseudo-distributed mode, Fully Distributed mode), Configuring XML files.							
UNIT-III (10 Hrs)	Writing Map Reduce Programs: A Weather Dataset, Understanding Hadoop API for Map Reduce Framework (Old and New), Basic programs of Hadoop Map Reduce: Driver code, Mapper code, Reducer code, Record Reader, Combiner, Partitioner							
UNIT-IV (10 Hrs)	Hadoop I/O: The Writable Interface, Writable Comparable and comparators, Writable Classes: Writable wrappers for Java primitives, Text, Bytes Writable, Null Writable, Object Writable and Generic Writable, Writable collections, Implementing a Custom Writable: Implementing a Raw Comparator for speed, Custom comparators							

UNIT-V (10 Hrs)	Pig: Hadoop Programming Made Easier, Admiring the Pig Architecture, Going with the Pig Latin Application Flow, Working through the ABCs of Pig Latin, Evaluating Local and Distributed Modes of Running Pig Scripts, Checking out the Pig Script Interfaces, Scripting with Pig Latin, Applying Structure to Hadoop Data with Hive: Saying Hello to Hive, Seeing How the Hive is Put Together, Getting Started with Apache Hive, Examining the Hive Clients, Working with Hive Data Types, Creating and Managing Databases and Tables, Seeing How the Hive Data Manipulation Language Works, Querying and Analyzing Data
Text Books:	
1.	Big Java 4th Edition, Cay Horstmann, Wiley John Wiley & Sons, INC
2.	Hadoop: The Definitive Guide by Tom White, 3rd Edition, O'Reilly
3.	Hadoop in Action by Chuck Lam, MANNING Publ
4.	Hadoop for Dummies by Dirk deRoos, Paul C.Zikopoulos, Roman B.Melnyk,Bruce Brown and Rafael Coss
Reference Books:	
1.	Hadoop in Practice by Alex Holmes, MANNING Publ
2.	HadoopMapReduce Cookbook, SrinathPerera, ThilinaGunarathne
Web Resources:	
1.	Hadoop: https://hadoop.apache.org/
2.	Hive: https://cwiki.apache.org/confluence/display/Hive/Home/
3.	Piglatin: https://pig.apache.org/docs/r0.7.0/tutorial.html

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1109	PE	3	0	0	3	25	75	3 Hrs.
APPLIED CRYPTOGRAPHY								
Program Elective -2								
Course Objectives:								
1.	Student learns the basic concepts of symmetric cryptography and simple encryption methods.							
2.	An understanding of the RSA cryptosystem, the mathematics used in the system, and the ability to encrypt and decrypt clear text using the system.							
3.	To learn the properties of message authentication codes and the ability to use hash functions to build a message authentication code.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Demonstrate the basics of Cryptographic protocols							K1
2.	Explain the concepts of Stream Ciphers and Public Key Encryption							K2
3.	Demonstrate Number Theory for Symmetric and Asymmetric Ciphers and discuss various Ciphers							K3
4.	Discuss Hashing Algorithms and Message Authentication Codes							K2
5.	Discuss Key-Exchange algorithms and Real world Implementations							K2
SYLLABUS								
UNIT-I (10 Hrs)	Foundations, Protocol Building Blocks, Basic Protocols, Advanced Protocols - Zero-Knowledge Proofs, Zero-Knowledge Proofs of Identity, Blind Signatures, Identity-Based Public- Key Cryptography, Key Length, Key Management, Electronic Codebook Mode, Block Replay, Cipher Block Chaining Mode, Stream Ciphers, Self-Synchronizing Stream Ciphers, Cipher- Feedback Mode, Synchronous Stream Ciphers, Output-Feedback Mode, Counter Mode, Other Block-Cipher Modes, Choosing a Cipher Mode.							
UNIT-II (8 Hrs)	Information Theory, Complexity Theory, Number Theory, Factoring, Prime Number Generation, Discrete Logarithms in a Finite Field, Data Encryption Standard (DES), IDEA, CAST, Blowfish, RC5, Double Encryption, Triple Encryption.							
UNIT-III (10 Hrs)	Pseudo-Random-Sequence Generators and Stream Ciphers- Linear Congruential Generators, Linear Feedback Shift Registers, Stream Ciphers using LFSRs, RC4, Feedback with Carry Shift Registers, Stream Ciphers Using FCSRs, Nonlinear-Feedback Shift Registers, Other Stream Ciphers, One-Way Hash Functions- MD5, Secure Hash Algorithm (SHA), One Way Hash Functions Using Symmetric Block, Using Public Key Algorithms, Message Authentication Codes.							

UNIT-IV (10 Hrs)	Public-Key Algorithms, Knapsack Algorithms, RSA, RabinmElGamal, Elliptic Curve Cryptosystems, Digital Signature Algorithm (DSA), DSA Variants, Gost Digital Signature Algorithm, Discrete Logarithm Signature Schemes, Ong-Schnorr-Shamir, Schnorr, Converting Identification Schemes to Signature Schemes.
UNIT-V (10 Hrs)	Diffie- Hellman, Station-to-Station Protocol, Multiple-Key Public-Key Cryptography, Subliminal Channel, Undeniable Digital Signatures, Designated Confirmer Signatures, Kerberos, Privacy-Enhanced Mail (PEM), Message Security Protocol (MSP), Pretty Good Privacy (PGP), Smart Cards, Public-Key Cryptography Standards (PKCS).
Text Books:	
1.	Applied Cryptography: Protocols, Algorithms, and Source Code in C, 2nd Edition, Bruce Schneier, John Wiley & Sons Inc, 1996
2.	Cryptography and Network Security, 6th Edition, William Stallings, Pearson Education, March 2013
Reference Books:	
1.	Modern Cryptography Theory and Practical, Wenbo Mao, Pearson Education, 2004
2.	Cryptography and network security, Behrouz A. Forouzan, McGraw-Hill, Inc., 2008

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1110	PE	3	0	0	3	25	75	3 Hrs.
ADVANCED COMPUTER NETWORKS								
Program Elective -2								
Course Objectives:								
1.	The course is aimed at providing basic understanding of Computer networks starting with OSI Reference Model, Protocols at different layers with special emphasis on IP, TCP & UDP and Routing algorithms							
2.	Some of the major topics which are included in this course are CSMA/CD, TCP/IPimplementation, LANs/WANs, internetworking technologies, Routing and Addressing.							
3.	Provide the mathematical background of routing protocols							
4.	Develop some familiarity with current research problems and research methods in advance computer networks.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Illustrate reference models with layers, protocols and interfaces.							K1
2.	Describe Sub netting and Addressing of IP V4andIPV6.							K2
3.	Describe and Analysis of basic protocols of computer networks, and how they can be used to assist in network design and implementation.							K2
4.	Discuss the concepts of congestion control and quality of service							K2
5.	Demonstrate Data Communications System and its components.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Network layer: Network Layer design issues: store-and forward packet switching, services provided transport layers, implementation connection less services, implementation connection oriented services, comparison of virtual –circuit and datagram subnets. Routing Algorithm –shortest path routing, flooding, distance vector routing, link state routing, Hierarchical routing, Broadcast routing, Multicasting routing, routing for mobiles Hosts, routing in Adhoc networks-congestion control algorithms-Load shedding, Congestion control in DatagramSubnet.							
UNIT-II (10 Hrs)	IPv4 Address: Address space, notations, classful addressing, classless addressing network addressing translation (NAT), IPv6 Address structure address space, Internetworking need for network layer internet as a data gram, internet as connection less network. IPv4 datagram, Fragmentation, checksum, options. IPv6 Advantages, packet format, extension Headers, Transition form IPV4 to IPV6							
UNIT-III (8 Hrs)	Process to process delivery: client/server paradigm, multiplexing and Demultiplexing, connectionless versus connection oriented services, reliable versus reliable.UDP: well known ports for UDP, user data gram, check sum, UDP operation, and uses of UDP.TCP:							

	TCP services, TCP features, segment, A TCP connection, Flow control, error control, congestion control. SCTP: SCTP services SCTP features, packet format, An SCTP association, flow control, error control.
UNIT-IV (10 Hrs)	Congestion control: open loop congestion control, closed loop congestion control, Congestion control in TCP, frame relay, QUALITY OF SERVICE: flow characteristics, flow classes TECHNIQUES TO IMPROVE QOS: scheduling, traffic shaping, resource reservation, admission control. Emerging trends Computer Networks: Motivation for mobile computing: protocol Stack Issues in Mobile Computing Environment, Mobility issues in mobile computing, data dissemination security issues mobile networks
UNIT-V (10 Hrs)	Domain name system: The name space, resource records, name servers E-mail: architecture and services, the user agent, message formats, message transfer, final delivery. Www: architecture overview, static web documents, dynamic web documents, hyper text transfer protocol, performance elements, the wireless web. Multimedia: introduction digital a audio , Audio compression, streaming audio, internet radio, voice over IP, introduction to video, video compression, video on demand, the MBone-the multicast back bone
Text Books:	
1.	Data communications and networking 4 th edition Behrouz A Fourzan, TMH
2.	Computer networks 4 th edition Andrew S Tanenbaum, Pearson
3.	Computer networks, Mayank Dave, CENGAGE
Reference Books:	
1.	Computer networks, A system Approach, 5 th ed, Larry L Peterson and Bruce S Davie, Elsevier
Weblinks:	
1.	http://nptel.iitm.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Computer%20networks/New_index1.html .
2.	http://nptel.iitm.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Computer%20networks/New_index1.html .

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1111	PE	3	0	0	3	25	75	3 Hrs.
EMBEDDED COMPUTING								
Program Elective -2								
Course Objectives:								
1.	To demonstrate the basic functions of the embeddedsystem.							
2.	To summarize the behavior of various software tools like GNU, GCC,gdb.							
3.	To infer GPS and GSM module interfacing							
4.	To develop client server models using TCP and UDP socketprogramming							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Describes the differences between the general computing system and the embedded computing system.							4
2.	Summarizes various software development tools like GNU, GCC etc.							2
3.	Develop interface modules for various types of sensors							3
4.	Write client server program using TCP and UDP sockets							3
SYLLABUS								
UNIT-I (10 Hrs)	Programming on Linux Platform: System Calls, Scheduling, Memory Allocation, Timers, Embedded Linux, Root File System, Busy Box. Operating System Overview: Processes, Tasks, Threads, Multi-Threading, Semaphore, Message Queue.							
UNIT-II (8 Hrs)	Introduction to Software Development Tools: GNU GCC, make, gdb, static and dynamic linking, C libraries, compiler options, code optimization switches, lint, code profiling tools.							
UNIT-III (10 Hrs)	Interfacing Modules: Sensor and actuator interface, data transfer and control, GPS, GSM module interfacing with data processing and display, OpenCV for machine vision, Audio signal processing.							
UNIT-IV (10 Hrs)	Networking Basics: Sockets, ports, UDP, TCP/IP, client server model, socket programming, 802.11, Bluetooth, ZigBee, SSH, firewalls, network security.							
UNIT-V (10 Hrs)	IA32 Instruction Set:application binary interface, exception and interrupt handling, interrupt latency, assemblers, assembler directives, macros, simulation and debugging tools.							

Text Books:

1.	Peter Barry and Patrick Crowley, “Modern Embedded Computing”, 1st Edition., Elsevier/Morgan Kaufmann, 2012.
2.	Linux Application Development – Michael K. Johnson, Erik W. Troan, Addison Wesley, 1998.
3.	Assembly Language for x86 Processors by Kip R. Irvine
4.	Intel® 64 and IA-32 Architectures Software Developer Manuals

Reference Books:

1.	Abraham Silberschatz, Peter B. Galvin and Greg Gagne, “Operating System Concepts”, Wiley
2.	Maurice J. Bach, “The Design of the UNIX Operating System”, Prentice-Hall
3.	W. Richard Stevens, “UNIX Network Programming”, Pearson

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1112	PE	3	0	0	3	25	75	3 Hrs.
PARALLEL COMPUTER ARCHITECTURE								
Program Elective -2								
Course Objectives:								
1.	To demonstrate the basic functions of the embedded system. To understand the principles of parallel computer architecture							
2.	To understand the design of parallel computer systems including modern parallel architectures							
3.	To assess the communication and computing possibilities of parallel system architecture and to predict the performance of parallel applications							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Students accustomed with the representation of data, addressing modes, and instructions sets.							K2
2.	Students able to understand parallelism both in terms of a single processor and multiple processors Technical knowhow of parallel hardware constructs to include instruction-level parallelism for multi core processor design							K3
SYLLABUS								
UNIT-I (8Hrs)	Fundamentals of Computer Design Defining Computer Architecture – Trends in Technology – Trends in Power in Integrated Circuits– Trends in Cost – Dependability – Measuring, Reporting and Summarizing Performance – Quantitative Principles of Computer Design – Basic and Intermediate concepts of pipelining – Pipeline Hazards – Pipelining Implementation issues.							
UNIT-II (10Hrs)	Instruction-Level Parallelism and Its Exploitation Instruction-Level Parallelism: Concepts and Challenges – Basic Compiler Techniques for Exposing ILP – Reducing Branch Costs with Prediction – Overcoming Data Hazards with Dynamic Scheduling – Dynamic Scheduling: Algorithm and Examples – Hardware-Based Speculation – Exploiting ILP Using Multiple Issue and Static Scheduling – Exploiting ILP Using Dynamic Scheduling, Multiple Issue and Speculation – Studies of the Limitations of ILP – Limitations on ILP for Realizable Processors – Hardware versus Software Speculation – Using ILP Support to Exploit Thread-Level Parallelism							
UNIT-III (10 Hrs)	Data-Level and Thread-Level Parallelism Vector Architecture – SIMD Instruction Set Extensions for Multimedia – Graphics Processing Units – Detecting and Enhancing Loop-Level Parallelism – Centralized Shared-Memory Architectures – Performance of Shared-Memory Multiprocessors – Distributed Shared Memory and Directory Based Coherence – Basics of Synchronization – Models of Memory Consistency – Programming Models and Workloads for Warehouse-Scale							

	Computers – Computer Architecture of Warehouse-Scale Computers – Physical Infrastructure and Costs of Warehouse-Scale Computers
UNIT-IV (10 Hrs)	Memory Hierarchy Design Cache Performance – Six Basic Cache Optimizations – Virtual Memory – Protection and Examples of Virtual Memory – Ten Advanced Optimizations of Cache Performance – Memory Technology and Optimizations – Protection: Virtual Memory and Virtual Machines – The Design of Memory Hierarchies
UNIT-V (10 Hrs)	Storage Systems & Case Studies Advanced Topics in Disk Storage – Definition and Examples of Real Faults and Failures – I/O Performance, Reliability Measures and Benchmarks – Designing and Evaluating an I/O System – The Internet Archive Cluster Case Studies / Lab Exercises: INTEL i3, i5, i7 processor cores, NVIDIA GPUs, AMD, ARM processor cores – Simulators – GEM5, CACTI, SIMICS, Multi2sim and INTEL Software development tools.
Text Books:	
1.	David.A.Patterson, John L.Hennessy, "Computer Architecture: A Quantitative approach", Elsevier, 5 th Edition 2012.
2.	K.Hwang, NareshJotwani, “Advanced Computer Architecture, Parallelism, Scalability, Programmability”, Tata McGraw Hill, 2 nd Edition 2010
Reference Books:	
1	V Rajaraman, Ram Murthy C.Siva “Parallel Computers: Architecture and Programming”, PHI

Code	Category	L	T	P	C	I.M	E.M	Exam
M19RD1101	RD	2	0	0	2	25	75	3 Hrs.
RESEARCH METHODOLOGY AND IPR								
(Common to CST,CS,PSA, IT & CAD)								
Course Objectives:								
1.	To bring awareness on Research Methodology and research ethics.							
2.	Familiarize the concepts of IPR.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Analyze research related information							K4
2.	Formulate a Research Proposals and Publish papers with research ethics							K6
3.	Award for Intellectual Property Rights like Patents, Trade and Copyrights							K5
4.	Analyze Various Intellectual Property Rights							K4
5.	AssessNew Developments of IPRs in National and International level							K5
SYLLABUS								
UNIT-I (6 Hrs)	Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations							
UNIT-II (6 Hrs)	Effective literature studies approaches, analysis Plagiarism, Research ethics, Effective technical writing, how to write report, Paper Developing a Research Proposal, Format of research proposal, a presentation and assessment by a review committee							
UNIT-III (6 Hrs)	Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, Patenting under PCT.							
UNIT-IV (4 Hrs)	Patent Rights: Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications.							
UNIT-V (6 Hrs)	New Developments in IPR: Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies, IPR and IITs.							
Text Books:								
1.	Stuart Melville and Wayne Goddard, —Research methodology: an introduction for science & engineering students’l							
2.	Wayne Goddard and Stuart Melville, —Research Methodology: An Introductionl							
3.	Ranjit Kumar, 2nd Edition, —Research Methodology: A Step by Step Guide for beginnersl							

Reference Books:	
1.	Halbert, —Resisting Intellectual Property, Taylor & Francis Ltd, 2007.
2.	Mayall, —Industrial Design, McGraw Hill, 1992.
3.	Niebel, —Product Design, McGraw Hill, 1974.
4.	Asimov, —Introduction to Design, Prentice Hall, 1962.
5.	Robert P. Merges, Peter S. Menell, Mark A. Lemley, — Intellectual Property in New Technological Age, 2016.
6.	T. Ramappa, —Intellectual Property Rights Under WTO, S. Chand, 2008

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1114	PC	0	0	4	2	25	75	3 Hrs.
ADVANCED DATA STRUCTURES LAB								
Course Objectives:								
1.	Understand and remember algorithms and their analysis procedure.							
2.	Introduce the concept of data structures through ADT including Dictionaries, List, Advanced Search Trees							
3.	To design and implement various data structure algorithms.							
4.	To introduce various techniques for representation of the data in the real world.							
5.	To develop application using data structure algorithms.							
6.	To Learn Various Text Processing Algorithms and Computational geometry methods.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Develop solutions for a range of problems using object oriented programming.							K3
2.	Implement complex problems using advanced data structures (like Dictionaries, Skip Lists and trees.).							K3
3.	Implement operations like searching, insertion, and deletion, traversing mechanism etc. on various data structures.							K3
4.	Utilize data structures in the applications such as binary search trees, AVL and B Trees							K3
SYLLABUS								
1.	Perform various operations on AVL Trees							
2.	Perform various operations on BST							
3.	Implementation of Static Hashing							
4.	Implementation of Huffman coding							
5.	Implementation of BTree.							
6.	Consider telephone book database of N clients. Make use of a hash table implementation to quickly look up client's telephonenumber.							
7.	Implement all the functions of a dictionary (ADT) using hashing. Data: Set of (key, value) pairs, Keys are mapped to values, Keys must be comparable, Keys must be unique Standard Operations: Insert(key, value), Find(key), Delete(key)							
8.	For given set of elements create skip list. Find the element in the set that is closest to some given value.							
9.	Implement KMP algorithm for Pattern Matching							

10.	Implement Boyer-Moore algorithm for PatternMatching
11.	Implement Naïve string matchingalgorithm.
12.	Implement insertion, deletion, display and search operation in m-way B tree (i.e. a non- leaf node can have atmost m children) for the given data as integers (Test the program for m=3, 5,7).
13.	Implementation ofSkiplists
Reference Books:	
1.	Mark Allen Weiss, Data Structures and Algorithm Analysis in C++, 4 th Edition, Person, 2013.
2.	Advanced Data Structures ReemaThareja,S.RamaSree, Oxford Higher Education
3.	Data Structures, Algorithms And Applications In C++ 2nd Edition Sartaj Sahni

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1115	PC	0	0	4	2	25	75	3 Hrs.
COMPUTING LAB								
Course Objectives:								
1.	To practice Python programming							
2.	To introduce to Planning, Natural Language Processing and Expert Systems.							
3.	Understand big data and Apache Hadoop Eco system							
4.	Discussion of various cryptographic techniques							
Course Outcomes : At the end of the coursestudents will be able to								
S.No	Outcome							Knowledge Level
1.	Write programs using different python packages.							K2
2.	Apply AI problem solving approaches to natural language processing. (Specialization 1)							K3
3.	Organize Real time data processing using Hadoop. (Specialization 2)							K3
4.	Develop different cryptography algorithms in java (Specialization 3)							K3
SYLLABUS								
Note: First SIX experiments are mandatory. Remaining experiments can be done based on the student’s choice of any one specialization.								
1.	a)	Write a python program to print the multiplication table for the given number?						
	b)	Write a python program to check whether the given number is prime or not?						
	c)	Write a python program to find factorial of the givennumber?						
2.	Write a python program to implement simple Chatbot?							
3.	a)	Write a python program to implement List operations (Nested List, Length, Concatenation, Membership, Iteration, Indexing and Slicing)?						
	b)	Write a python program to implement List methods (Add, Append, Extend& Delete).						
4.	a)	Write a python program to Illustrate Different SetOperations?						
	b)	Write a python program to generate Calendar for the given month and year?						
	c)	Write a python program to implement Simple Calculatorprogram?						
5.	a)	Write a python program to Add Two Matrices.						
	b)	Write a python program to Transpose a Matrix.						

6.	a)	Write a python program to remove punctuations from the given string?
	b)	Write a python program to sort the sentence in alphabetical order?
Artificial Intelligence Specialization:		
1.		Write a python program to implement Breadth First Search Traversal?
2.		Write a python program to implement Water Jug Problem?
3.		Write a program to implement Hangman game using python.
4.		Write a program to implement Tic-Tac-Toe game
5.	a)	Write a python program to remove stop words for a given passage from a text file using NLTK?
	b)	Write a python program to implement stemming for a given sentence using NLTK?
	c)	Write a python program to POS (Parts of Speech) tagging for the give sentence using NLTK?
Big Data Specialization:		
1.	a)	Perform setting up and Installing Hadoop in its three operating modes: Standalone, Pseudo distributed, Fully distributed
	b)	Use web based tools to monitor your Hadoop setup.
2.		Implement the following file management tasks inHadoop: • Adding files anddirectories • Retrievingfiles • Deletingfiles
3.		Run a basic Word Count Map Reduce program to understand Map ReduceParadigm.
4.		Write a Map Reduce program that mines weatherdata. Weather sensors collecting data every hour at many locations across the globe gather a large volume of log data, which is a good candidate for analysis with MapReduce, since it is semi structured and record-oriented.
5.		Implement Matrix Multiplication with Hadoop Map Reduce
6.		Install and Run Pig then write Pig Latin scripts to sort, group, join, project, and filter Yourdata.
7.		Install and Run Hive then use Hive to create, alter, and drop databases, tables, views, functions, andindexes

Cryptography Specialization:	
1.	Exercise –1: Write a Java program to perform encryption and decryption using the following algorithms: a) Ceaser Cipher b) Substitution Cipher c) Hill Cipher
2.	Exercise – 2: Write a Java program to implement the 3 DES and AES algorithms.
3.	Exercise – 3: Write a JAVA program to implement the BlowFish algorithm
4.	Exercise-4 : Implement MD-5 using Java
5.	Exercise-5: Write a Java program to implement RSA (2048 Key Length) Algorithm.
6.	Exercise-6: Implement the Diffie-Hellman Key Exchange mechanism using HTML and JavaScript. Consider the end user as one of the parties (Alice) and the JavaScript application as other party (bob).
7.	Exercise-7: Calculate the message digest of a text using the SHA-2 algorithm in JAVA.
Reference Books:	
1.	Let us Python, YashavantKanetkar BPB publications
2.	Artificial Intelligence with Python: A Comprehensive Guide to Building Intelligent Apps for Python Beginners and Developers, Prateek Joshi Packt Publishing
3.	Hadoop Map Reduce Cookbook, SrinathPerera&ThilinaGunarathne, 2013, PACKT PUBLISHING.
4.	Beginning Cryptography with Java, David Hook Wiley India Private Limited

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0001	AC	2	0	0	0	0	0	--
ENGLISH FOR RESEARCH PAPER WRITING								
Course Objectives:								
1.	Understand how to improve your writing skills and level of readability							
2.	Learn about what to write in each section.							
3.	Understand the skills needed when writing a Title.							
4.	Ensure the good quality of paper at very first-time submission							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand that how to improve your writing skills and level of readability							K2
2.	Learn about what to write in each section							K2
3.	Understand the skills needed when writing a Title Ensure the good quality of paper at very first time submission							K2
SYLLABUS								
UNIT-I (4Hrs)	Planning and Preparation, Word Order, Breaking up long sentences, Structuring Paragraphs and Sentences, Being Concise and Removing Redundancy, Avoiding Ambiguity and Vagueness							
UNIT-II (4Hrs)	Clarifying Who Did What, Highlighting Your Findings, Hedging And Criticizing, Paraphrasing and Plagiarism, Sections of a Paper.							
UNIT-III (4Hrs)	Abstracts,Introduction,Review of the Literature, Methods, Results, Discussion, Conclusions, The Final Check.							
UNIT-IV (4Hrs)	Key skills are needed when writing a Title, key skills are needed when writing an abstract, key skills are needed when writing an introduction, skills needed when writing a Review of the Literature, skills are needed when writing the Methods.							
UNIT-V (4Hrs)	skills needed when writing the Results, skills are needed when writing the Discussion, skills are needed when writing the Conclusions, useful phrases, how to ensure paper is as good as it could possibly be the first- time submission							
Text Books:								
1.	Goldbort R (2006) Writing for Science, Yale University Press (available on Google Books)							
2.	Day R (2006) How to Write and Publish a Scientific Paper, Cambridge University Press							
Reference Books:								
1.	Highman N (1998), Handbook of Writing for the Mathematical Sciences, SIAM. Highman's book.							
2.	Adrian Wallwork , English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0002	AC	2	0	0	0	0	0	--
DISASTER MANAGEMENT								
Course Objectives:								
1.	Learn to demonstrate a critical understanding of key concepts in disaster risk reduction and humanitarian response.							
2.	Critically evaluate disaster risk reduction and humanitarian response policy and practice from multiple perspectives.							
3.	Develop an understanding of standards of humanitarian response and practical relevance in specific types of disasters and conflict situations.							
4.	Critically understand the strengths and weaknesses of disaster management approaches, planning & programming in different countries, particularly their home country or the countries they work in.							
SYLLABUS								
UNIT-I (4Hrs)	Disaster: Definition, Factors And Significance; Difference Between Hazard And Disaster; Natural And Manmade Disasters: Difference, Nature, Types And Magnitude.							
UNIT-II (4Hrs)	Economic Damage, Loss Of Human And Animal Life, Destruction Of Ecosystem. Natural Disasters: Earthquakes, Volcanisms, Cyclones, Tsunamis, Floods, Droughts And Famines, Landslides And Avalanches, Man-made disaster: Nuclear Reactor Meltdown, Industrial Accidents, Oil Slicks And Spills, Outbreaks Of Disease And Epidemics, War And Conflicts							
UNIT-III (4Hrs)	Disaster Prone Areas In India: Study Of Seismic Zones; Areas Prone To Floods And Droughts, Landslides And Avalanches; Areas Prone To Cyclonic And Coastal Hazards With Special Reference To Tsunami; Post-Disaster Diseases And Epidemics.							
UNIT-IV (4Hrs)	Disaster Preparedness And Management Preparedness: Monitoring Of Phenomena Triggering A Disaster Or Hazard; Evaluation Of Risk: Application Of Remote Sensing, Data From Meteorological And Other Agencies, Media Reports: Governmental And Community Preparedness.							
UNIT-V (4Hrs)	Risk Assessment Disaster Risk: Concept And Elements, Disaster Risk Reduction, Global And National Disaster Risk Situation. Techniques Of Risk Assessment, Global Co-Operation In Risk Assessment And Warning, People's Participation In Risk Assessment. Strategies for Survival. Disaster Mitigation Meaning, Concept And Strategies Of Disaster Mitigation, Emerging Trends In Mitigation. Structural Mitigation And Non-Structural Mitigation, Programs Of Disaster Mitigation In India.							
Text Books:								
1.	R. Nishith, Singh AK, Disaster Management in India: Perspectives, issues and strategies New Royal Book Company.							

2.	Sahni, PardeepEt.Al. (Eds.),Disaster Mitigation Experiences And Reflections, Prentice Hall Of India, New Delhi.
3.	Goel S. L. Disaster Administration And Management Text And Case Studies ,Deep &Deep Publication Pvt. Ltd., New Delhi.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0003	AC	2	0	0	0	0	0	--
SANSKRIT FOR TECHNICAL KNOWLEDGE								
Course Objectives:								
1.	To get a working knowledge in illustrious Sanskrit, the scientific language in the world							
2.	Learning of Sanskrit to improve brain functioning							
3.	Learning of Sanskrit to develop the logic in mathematics, science & other subjects enhancing the memory power							
4.	The engineering scholars equipped with Sanskrit will be able to explore the huge knowledge from ancient literature							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understanding basic Sanskrit language.							K2
2	Ancient Sanskrit literature about science & technology can be understood.							K2
3	Being a logical language will help to develop logic in students.							K6
SYLLABUS								
UNIT-I (7Hrs)	Alphabets in Sanskrit, Past/Present/Future Tense, Simple Sentences							
UNIT-II (7Hrs)	Order, Introduction of roots, Technical information about Sanskrit Literature							
UNIT-III (7Hrs)	Technical concepts of Engineering-Electrical, Mechanical, Architecture, Mathematics							
Text Books:								
1.	“AbhyasaPustakam” – Dr.Vishwas, Samskrita-Bharati Publication, New Delhi							
2.	“Teach Yourself Sanskrit” PrathamaDiksha-VempatiKutumbasastry, Rashtriya Sanskrit Sansthanam, New Delhi Publication							
3.	“India’s Glorious Scientific Tradition” Suresh Soni, Ocean books (P) Ltd., New Delhi							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0004	AC	2	0	0	0	0	0	--
VALUE EDUCATION								
Course Objectives:								
1.	Understand the value of education and self- development							
2.	Imbibe good values in students							
3.	Let should know about the importance of character							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Knowledge of self-development							K1
2	Learn the importance of Human values							K2
3	Developing the overall personality							K3
SYLLABUS								
UNIT-I (4Hrs)	Values and self-development –Social values and individual attitudes. Work ethics, Indian vision of humanism.Moral and non- moral valuation. Standards and principles. Value judgements.							
UNIT-II (4Hrs)	Importance of cultivation of values. Sense of duty. Devotion, Self-reliance. Confidence, Concentration. Truthfulness, Cleanliness. Honesty, Humanity. Power of faith, National Unity, Patriotism, Love for nature ,Discipline							
UNIT-III (4Hrs)	Personality and Behavior Development - Soul and Scientific attitude. Positive Thinking. Integrity and discipline. Punctuality, Love and Kindness. Avoid fault Thinking. Free from anger, Dignity of labour. Universal brotherhood and religious tolerance. True friendship. Happiness Vs suffering, love for truth. Aware of self-destructive habits. Association and Cooperation, Doing best for saving nature							
UNIT-IV (4Hrs)	Character and Competence –Holy books vs Blind faith.Self-management and Good health. Science of reincarnation. Equality, Nonviolence ,Humility, Role of Women. All religions and same message. Mind your Mind, Self-control. Honesty, Studying effectively.							
Text Books:								
1.	Chakroborty, S.K. “Values and Ethics for organizations Theory and practice”, Oxford University Press, New Delhi							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0005	AC	2	0	0	0	0	0	--
CONSTITUTION OF INDIA								
Course Objectives:								
1.	Understand the premises informing the twin themes of liberty and freedom from a civil rights perspective.							
2.	To address the growth of Indian opinion regarding modern Indian intellectuals’ constitutional role and entitlement to civil and economic rights as well as the emergence of nationhood in the early years of Indian nationalism							
3.	To address the role of socialism in India after the commencement of the Bolshevik Revolution in 1917 and its impact on the initial drafting of the Indian Constitution							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Discuss the growth of the demand for civil rights in India for the bulk of Indians before the arrival of Gandhi in Indian politics.							K2
2	Discuss the intellectual origins of the framework of argument that informed the conceptualization of social reforms leading to revolution in India.							K2
3	Discuss the circumstances surrounding the foundation of the Congress Socialist Party [CSP] under the leadership of Jawaharlal Nehru and the eventual failure of the proposal of direct elections through adult suffrage in the Indian Constitution.							K2
4	Discuss the passage of the Hindu Code Bill of 1956.							K2
SYLLABUS								
UNIT-I (4Hrs)	History of Making of the Indian Constitution: History , Drafting Committee, (Composition & Working)							
UNIT-II (4Hrs)	Philosophy of the Indian Constitution: Preamble ,Salient Features							
UNIT-III (4Hrs)	Fundamental Rights,Right to Equality, Right to Freedom,Right against Exploitation, Right to Freedom of Religion,Cultural and Educational Rights,Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties							
UNIT-IV (4Hrs)	Organs of Governance: Parliament, Composition, Qualifications and Disqualifications ,Powers and Functions, Executive, President , Governor, Council of Ministers, Judiciary, Appointment and Transfer of Judges, Qualifications, Powers and Functions Local Administration: District’s Administration head: Role and Importance, Municipalities: Introduction, Mayor and role of Elected Representative, CEO of Municipal Corporation, Pachayati raj: Introduction, PRI: Zilla Panchayat. Elected officials and their roles, CEO Zilla Panchayat:							

	Position and role. Block level: Organizational Hierarchy (Different departments), Village level: Role of Elected and Appointed officials, Importance of grass root democracy
UNIT-V (4Hrs)	Election Commission: Election Commission: Role and Functioning. Chief Election Commissioner and Election Commissioners. State Election Commission: Role and Functioning. Institute and Bodies for the welfare of SC/ST/OBC and women.
Text Books:	
1.	The Constitution of India, 1950 (Bare Act), Government Publication.
2.	Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, 2015.
3.	M. P. Jain, Indian Constitution Law, 7th Edn., LexisNexis, 2014.
4.	D.D. Basu, Introduction to the Constitution of India, LexisNexis, 2015.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0006	AC	2	0	0	0	0	0	--
PEDAGOGY STUDIES								
SYLLABUS								
UNIT-I (4Hrs)	Introduction and Methodology: Aims and rationale, Policy background, Conceptual framework and terminology Theories of learning, Curriculum, Teacher education. Conceptual framework, Research questions. Overview of methodology and Searching.							
UNIT-II (4Hrs)	Thematic overview: Pedagogical practices are being used by teachers in formal and informal classrooms in developing countries, Curriculum, Teacher education.							
UNIT-III (4Hrs)	Evidence on the effectiveness of pedagogical practices, Methodology for the in depth stage: quality assessment of included studies. How can teacher education (curriculum and practicum) and the school curriculum and guidance materials best support effective pedagogy.							
UNIT-IV (4Hrs)	Theory of change, Strength and nature of the body of evidence for effective pedagogical practices Pedagogic theory and pedagogical approaches. Teachers’ attitudes and beliefs and Pedagogic strategies. Professional development: alignment with classroom practices and follow-up support							
UNIT-V (4Hrs)	Research gaps and future directions: Research design, Contexts, Pedagogy, Teacher education, Curriculum and assessment, Dissemination and research impact.							
Text Books:								
1.	Ackers J, Hardman F (2001) Classroom interaction in Kenyan primary schools, Compare, 31 (2): 245-261.							
2.	Agrawal M (2004) Curricular reform in schools: The importance of evaluation, Journal of Curriculum Studies, 36 (3): 361-379.							
3.	Akyeampong K (2003) Teacher training in Ghana - does it count? Multi-site teacher education research project (MUSTER) country report 1. London: DFID.							
4.	Akyeampong K, Lussier K, Pryor J, Westbrook J (2013) Improving teaching and learning of basic maths and reading in Africa: Does teacher preparation count? International Journal of Educational Development, 33 (3): 272–282.							
5.	Alexander RJ (2001) Culture and pedagogy: International comparisons in primary education. Oxford and Boston: Blackwell.							
6.	Chavan M (2003) Read India: A mass scale, rapid, „learning to read“ campaign.							
7.	www.pratham.org/images/resource%20working%20paper%202.pdf.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0007	AC	2	0	0	0	0	0	--
STRESS MANAGEMENT BY YOGA								
Course Objectives:								
1.	To achieve overall health of body and mind							
2.	To overcome stress							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Develop a healthy mind in a healthy body thus improving social health also.							K2
2	Improve efficiency							K2
SYLLABUS								
UNIT-I (7Hrs)	Definitions of Eight parts of yoga (Ashtanga)							
UNIT-II (7Hrs)	Yam and Niyam. Do's and Don'ts in life. i) Ahinsa, satya, astheya, brahmacharya and aparigraha ii) Shaucha, santosh, tapa, swadhyay, ishvarapranidhana							
UNIT-III (7Hrs)	Asan and Pranayam i) Various yog poses and their benefits for mind & body ii)Regularization of breathing techniques and its effects-Types pranayama							
Text Books:								
1.	‘Yogic Asanas for Group Training-Part-I’ : Janardan Swami YogabhyasiMandal, Nagpur							
2.	“Raja Yoga or conquering the Internal Nature” by Swami Vivekananda, AdvaitaAshrama (Publication Department), Kolkata							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0008	AC	2	0	0	0	0	0	--
PERSONALITY DEVELOPMENT THROUGH LIFEENLIGHTENMENT SKILLS								
Course Objectives:								
1.	To learn to achieve the highest goal happily							
2.	To become a person with a stable mind, pleasing personality and determination							
3.	To awaken wisdom in students							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Study of Shrimad-Bhagwad-Geeta will help the student in developing his personality and achieve the highest goal in life.							K2
2	The person who has studied Geeta will lead the nation and mankind to peace and prosperity.							K2
3	Study of Neetishatakam will help in developing versatile personality of students.							K2
SYLLABUS								
UNIT-I (7Hrs)	Neetisatakam-Holistic development of personality Verses- 19,20,21,22 (wisdom), Verses- 29,31,32 (pride & heroism) Verses- 26,28,63,65 (virtue),Verses- 52,53,59 (don'ts),Verses- 71,73,75,78 (do's)							
UNIT-II (7Hrs)	Approach to day to day work and duties. ShrimadBhagwadGeeta : Chapter 2-Verses: 41, 47,48 Chapter 3-Verses 13, 21, 27, 35, Chapter 6-Verses 5,13,17, 23, 35, Chapter 18-Verses 45, 46, 48.							
UNIT-III (7Hrs)	Statements of basic knowledge, ShrimadBhagwadGeeta: Chapter2-Verses 56, 62, 68, Chapter 12 -Verses 13, 14, 15, 16,17, 18, Personality of Role model. ShrimadBhagwadGeeta: Chapter2-Verses 17, Chapter 3-Verses 36,37,42, Chapter 4-Verses 18, 38,39, Chapter18 – Verses 37,38,63							
Text Books:								
1.	‘Srimad Bhagavad Gita” by Swami SwarupanandaAdvaita Ashram (Publication Department), Kolkata							
2.	Bhartrihari’s Three Satakam (Niti-sringar-vairagya) by P.Gopinath,Rashtriya Sanskrit Sansthanam, New Delhi.							



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION (Regulation R19) **M.TECH (COMPUTER SCIENCE TECHNOLOGY)** **DEPARTMENT OF COMPUTER SCIENCE TECHNOLOGY** (With effect from 2019-2020 Admitted Batch onwards) **II-SEMESTER**

Subject Code.	Name of the Subject	Category.	C	L	T	P	Int. Marks	External Marks	Total Marks
M19 CST 1201	Advanced Algorithms	PC	3	3	0	0	25	75	100
M19 CST 1202	Machine Learning	PC	3	3	0	0	25	75	100
#PE-III	Program Elective-III	PE	3	3	0	0	25	75	100
#PE-IV	Program Elective-IV	PE	3	3	0	0	25	75	100
M19 CST 1213	Advance Algorithm Lab	PC	2	0	0	4	25	75	100
M19 CST 1214	Machine Learning Lab	PC	2	0	0	4	25	75	100
M19 CST 1215	Mini Project with Seminar	MP	2	0	0	4	100	--	100
#AC-2	Audit Course-2*	AC	0	2	0	0	0	0	--
Total			18	14	0	12	250	450	700

	Code	Course
#PE-III	M19CST1203	Soft Computing
	M19CST1204	Advanced Network Principles and Protocols
	M19CST1205	Internet of Things
	M19CST1206	Open Source Programming
	M19CST1207	Pattern Recognition
#PE-IV	M19CST1208	Natural Language Processing
	M19CST1209	Full Stack Technologies
	M19CST1210	Parallel Algorithms
	M19CST1211	Object Oriented Software Engineering
	M19CST1212	Distributed Database

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1201	PC	3	0	0	3	25	75	3 Hrs.
ADVANCED ALGORITHMS								
Course Objectives:								
1.	Introduce students to the advanced methods of designing and analyzing algorithms.							
2.	The student should be able to choose appropriate algorithms and use it for a specific problem.							
3.	To familiarize students with basic paradigms and data structures used to solve advanced algorithmic problems.							
4.	Students should be able to understand different classes of problems concerning their computation difficulties.							
5.	To introduce the students to recent developments in the area of algorithmic design.							
6.	Introduce students to the advanced methods of designing and analyzing algorithms.							
Course Outcomes After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
1.	Analyze the complexity/performance of different algorithms.							K4
2.	Determine the appropriate data structure for solving a particular set of problems.							K3
3.	Categorize the different problems in various classes according to their complexity.							K3
4.	Students should have an insight of recent activities in the field of the advanced data structure.							K3
SYLLABUS								
UNIT-I (10Hrs)	Sorting: Review of various sorting algorithms, topological sorting Graph: Definitions and Elementary Algorithms: Shortest path by BFS, shortest path in edge-weighted case (Dijkasra's), depth-first search and computation of strongly connected components, emphasis on correctness proof of the algorithm and time/space analysis, example of amortized analysis.							
UNIT-II (10Hrs)	Matroids: Introduction to greedy paradigm, algorithm to compute a maximum weight maximal independent set. Application to MST. Graph Matching : Algorithm to compute maximum matching. Characterization of maximum matching by augmenting paths, Edmond's Blossom algorithm to compute augmenting path.							
UNIT-III (10Hrs)	Flow-Networks: Maxflow-mincut theorem, Ford-Fulkerson Method to compute maximum flow, Edmond-Karp maximum-flow algorithm. Matrix Computations: Strassen's algorithm and introduction to divide and conquer paradigm, inverse of a triangular matrix, relation between the time complexities of basic matrix operations, LUP-decomposition.							
UNIT-IV (10Hrs)	Shortest Path in Graphs: Floyd-Warshall algorithm and introduction to dynamic programming paradigm. More examples of dynamic programming. Modulo Representation of integers/polynomials: Chinese Remainder Theorem, Conversion between base							

	representation and modulo-representation. Extension to polynomials. Application: Interpolation problem. Discrete Fourier Transform (DFT): In complex field, DFT in modulo ring. Fast Fourier Transform algorithm. Schonhage-Strassen Integer Multiplication algorithm
UNIT-V (10Hrs)	Linear Programming: Geometry of the feasibility region and Simplex algorithm. NP-completeness: Examples, proof of NP-hardness and NP-completeness. One or more of the following topics based on time and interest Approximation algorithms, Randomized Algorithms, Interior Point Method, Advanced Number Theoretic Algorithm Recent Trends in problem solving paradigms using recent searching and sorting techniques by applying recently proposed data structures.
Text Books:	
1.	"Introduction to Algorithms" by Cormen, Leiserson, Rivest, Stein, MIT Press
2.	"The Design and Analysis of Computer Algorithms" by Aho, Hopcroft, Ullman, Pearson Education.
Reference Books:	
1.	"Algorithm Design" by Kleinberg and Tardos, Pearson Education

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1202	PC	3	0	0	3	25	75	3 Hrs.
MACHINE LEARNING								
Course Objectives:								
1.	To introduce students to the basic concepts and techniques of Machine Learning.							
2.	To become familiar with regression methods, classification methods, clustering methods.							
3.	To become familiar with the concepts of artificial neural networks.							
Course Outcomes After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
1.	Recognize the characteristics of machine learning algorithms and their applications to real world problems							K1
2.	Able to write and evaluate hypothesis							K2
3.	Apply kernel methods to solve real world problems.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction : Towards Intelligent Machines, Well posed Problems, Example of Applications in diverse fields, Data Representation, Domain Knowledge for Productive use of Machine Learning, Diversity of Data: Structured / Unstructured, Forms of Learning, Machine Learning and Data Mining, Basic Linear Algebra in Machine Learning Techniques.							
UNIT-II (10Hrs)	Supervised Learning : Rationale and Basics: Learning from Observations, Bias and Why Learning Works: Computational Learning Theory, Occam's Razor Principle and Over fitting Avoidance Heuristic Search in inductive Learning, Estimating Generalization Errors, Metrics for assessing regression, Metris for assessing classification.							
UNIT-III (10Hrs)	Statistical Learning : Machine Learning and Inferential Statistical Analysis, Descriptive Statistics in learning techniques, Bayesian Reasoning: A probabilistic approach to inference, K-Nearest Neighbor Classifier. Discriminant functions and regression functions, Linear Regression with Least Square Error Criterion, Logistic Regression for Classification Tasks, Fisher's Linear Discriminant and Thresholding for Classification, Minimum Description Length Principle.							
UNIT-IV (10Hrs)	Support Vector Machines (SVM) : Introduction, Linear Discriminant Functions for Binary Classification, Perceptron Algorithm, Large Margin Classifier for linearly seperable data, Linear Soft Margin Classifier for Overlapping Classes, Kernel Induced Feature Spaces, Nonlinear Classifier, Regression by Support vector Machines. Learning with Neural Networks : Towards Cognitive Machine, Neuron Models, Network Architectures, Perceptrons, Linear neuron and the Widrow-Hoff Learning Rule, The error correction delta rule.							

UNIT-V (10Hrs)	Multilayer Perceptron Networks and error back propagation algorithm, Radial Basis Functions Networks. Decision Tree Learning: Introduction, Example of classification decision tree, measures of impurity for evaluating splits in decision trees, ID3, C4.5, and CART decision trees, pruning the tree, strengths and weakness of decision tree approach.
Text Books:	
1.	Applied Machine Learning, M. Gopal, McGraw Hill Education
2.	Machine Learning: A Probabilistic Perspective, Kevin Murphy, MIT Press, 2012
Reference Books:	
1.	Pattern Recognition and Machine Learning, Christopher Bishop, Springer, 2007
2.	Programming Collective Intelligence: Building Smart Web 2.0 Applications - Toby Segaran
3.	Building Machine Learning Systems with Python - Willi Richert, Luis Pedro Coelho
4.	The Elements of Statistical Learning, Trevor Hastie, Robert Tibshirani, Jerome Friedman, Springer 2009 (freely available online)

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1203	PE	3	0	0	3	25	75	3 Hrs.
SOFT COMPUTING								
(Program Elective-III)								
Course Objectives:								
1.	Develop the skills to gain a basic understanding of neural network theory and fuzzy logic theory.							
2.	Introduce students to artificial neural networks and fuzzy theory from an engineering perspective.							
Course Outcomes At the end of the course, student will be able to								
S.No	Outcome							Knowledge Level
1.	Comprehend the fuzzy logic and the concept of fuzziness involved in various systems and fuzzy set theory							K3
2.	Understand the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic							K2
3.	To understand the fundamental theory and concepts of neural networks, Identify different neural network architectures, algorithms, applications and their limitations							K2
4.	Understand appropriate learning paradigms and its applications rules for each of the architectures and learn several neural network							K2
5.	Reveal different applications of these models to solve engineering and other problems.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Fuzzy Set Theory: Introduction to Neuro – Fuzzy and Soft Computing, Fuzzy Sets, Basic Definition and Terminology, Set-theoretic Member Function Formulation and Parameterization, Fuzzy Rules and Fuzzy Reasoning, Extension Principle and Fuzzy Relations Fuzzy If-Then Rules, Fuzzy Reasoning, Fuzzy Inference Systems, Mamdani Fuzzy Models, Surgeon Fuzzy Models, Tsukamoto Fuzzy Models, Input Space Partitioning and Fuzzy Modeling.							
UNIT-II (10 Hrs)	Optimization: Derivative based Optimization, Descent Methods, The Method of Steepest Descent, Classical Newton’s Method, Step Size Determination, Derivative-free Optimization, Genetic Algorithms, Simulated Annealing and Random Search – Downhill Simplex Search.							
UNIT-III (10 Hrs)	Artificial Intelligence: Introduction, Knowledge Representation, Reasoning, Issues and Acquisition: Propositional and Predicate Calculus Rule Based knowledge Representation Symbolic Reasoning under Uncertainty Basic knowledge Representation Issues Knowledge acquisition, Heuristic Search: Techniques for Heuristic search Heuristic Classification State Space Search: Strategies Implementation of Graph Search based on Recursion Patent directed Search Production System and Learning.							
UNIT-IV (10 Hrs)	Neuro Fuzzy Modeling: Adaptive Neuro-Fuzzy Inference Systems, Architecture – Hybrid Learning Algorithm, Learning Methods that Cross-fertilize ANFIS and RBFN – Coactive							

	Neuro Fuzzy Modeling, Framework Neuron Functions for Adaptive Networks – Neuro Fuzzy Spectrum.
UNIT-V (10 Hrs)	Applications Of Computational Intelligence: Printed Character Recognition, Inverse Kinematics Problems, Automobile Fuel Efficiency Prediction, Soft Computing for Color Recipe Prediction.
Text Books:	
1	J.S.R.Jang, C.T.Sun and E.Mizutani, “Neuro-Fuzzy and Soft Computing”, PHI, 2004, Pearson Education 2004.
2	N.P.Padhy, “Artificial Intelligence and Intelligent Systems”, Oxford University Press, 2006.
Reference Books:	
1	Elaine Rich & Kevin Knight, Artificial Intelligence, Second Edition, Tata Mcgraw Hill Publishing Comp., 2006, New Delhi.
2	Timothy J.Ross, “Fuzzy Logic with Engineering Applications”, McGraw-Hill, 1997.
3	Davis E.Goldberg, “Genetic Algorithms: Search, Optimization and Machine Learning”, Addison Wesley, N.Y., 1989

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1204	PE	3	0	0	3	25	75	3 Hrs.
ADVANCED NETWORK PRINCIPLES AND PROTOCOLS								
(Program Elective-III)								
Course Objectives:								
1	Understand the architecture of the Internet protocols as a layered model							
2	To understand the fundamentals of data transmission, encoding and multiplexing							
3	To understand how the various components of wide area networks and local area networks work together							
Course Outcomes At the end of the course, student will be able to								
S.No	Outcome							Knowledge Level
1	Familiarization of the different layers of TCP/IP protocol stack							K3
2	Understanding of the working principle of different protocols at different layers							K2
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Networks - Application of Networks - Architecture Topology Switching - SLIP, PPP -ALOHA protocols, CSMA/CD, IEEE 802.3, 802.4, 802.5							
UNIT-II (10 Hrs)	Network Layer Issues- Routing, Congestion control- Internetworking - Issues, Address Learning Bridges, Spanning tree, Source routing, Bridges, Routers, Gateway.							
UNIT-III (10 Hrs)	Network Protocol- IP datagram - hop by hop routing, ARP, RARP, DHCP -Sub net Addressing, Address Masking, ICMP, RIP, RIPv2, OSPF, DNS, LAN and WAN Multicast.							
UNIT-IV (10 Hrs)	Transport Layer- Design issues, Connection Management, Transmission Control Protocol (TCP) - User Datagram Protocol (UDP).							
UNIT-V (10 Hrs)	Application Layer Protocol- Telnet - TFTP - FTP - SMTP - Ping Finger, Bootstrap Network Time Protocol-SNMP.							
Text Books:								
1.	Andrew S. Tanenbaum and David J. Wetherall, “Computer Networks”, 5th Edition, Pearson, 2011							
2.	William Stallings, "Data and Computer Communications”, 9th Edition, Pearson, 2011							
Reference Books:								
1.	W Richard Stevens and G. Gabrani, "TCP/IP Illustrated - Volume I, The protocols", Pearson Education, 2009							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1205	PC	3	0	0	3	25	75	3 Hrs.
INTERNET OF THINGS								
(Program Elective-III)								
Course Objectives:								
1.	To Understand Smart Objects and IoT Architectures.							
2.	To learn about various IOT-related protocols							
3.	To build simple IoT Systems using Arduino and Raspberry Pi.							
4.	To understand data analytics and cloud in the context of IoT							
5.	To develop IoT infrastructure for popular applications.							
Course Outcomes: After completion of course, students will be able to:								
S.No	Outcome							Knowledge Level
1.	Summarize on the term 'internet of things' in different contexts.							K2
2.	Analyze various protocols for IoT.							K4
3.	Design a PoC of an IoT system using Rasperry Pi/Arduino							K6
4.	Apply data analytics and use cloud offerings related to IoT.							K3
5.	Analyze applications of IoT in real time scenario							K4
SYLLABUS								
UNIT-I (10 Hrs)	FUNDAMENTALS OF IoT: Evolution of Internet of Things, Enabling Technologies, IoTArchitectures,oneM2M, IoT World Forum (IoTWF) and Alternative IoT models, Simplified IoT Architecture andCore IoT Functional Stack, Fog, Edge and Cloud in IoT, Functional blocks of an IoT ecosystem, Sensors,Actuators, Smart Objects and Connecting Smart Objects							
UNIT-II (10 Hrs)	IoT PROTOCOLS: IT Access Technologies: Physical and MAC layers, topology and Security of IEEE 802.15.4, 802.15.4g, 802.15.4e, 1901.2a, 802.11ah and Lora WAN, Network Layer: IP versions, Constrained Nodes andConstrained Networks, Optimizing IP for IoT: From 6LoWPAN to 6Lo, Routing over Low Power and Lossy Networks, Application Transport Methods: Supervisory Control and Data Acquisition, Application LayerProtocols: CoAP and MQTT. Bluetooth Smart Connectivity-Overview, Key Versions, BLE-Bluetooth Low EnergyProtocol, Low Energy Architecture.							
UNIT-III (10 Hrs)	Data Acquiring and storage, organizing the Data and Analytics, Cloud Computing paradigm for Data Collection, Storage and computing.DESIGN AND DEVELOPMENT: Embedded computing logic, Microcontroller, Systemon Chips, IoT system building blocks, Arduino, Board details, IDE programming, Raspberry Pi.							
UNIT-IV (10 Hrs)	Arm Based Embedded System Design: ARM Cortex-A class processor, Embedded Devices-ARM Cortex-MClass processor, Networking-Bluetooth Smart Technology Introduction to embedded systems: CPUs vs MCU’s vs Embedded Systems, Examples, Options for BuildingEmbedded Systems, Features of Embedded Systems, Building Embedded Systems, Building Embedded Systemsusing MCUs, Introduction to mbedTM Platform							

UNIT-V (10 Hrs)	CASE STUDIES/INDUSTRIAL APPLICATIONS: Cisco IoT system, IBM Watson IoT platform, Manufacturing, Converged Plant wide Ethernet Model (CPwE), Power Utility Industry, GridBlocks Reference Model, Smart and Connected Cities: Layered architecture, Smart Lighting, Smart Parking Architecture and Smart Traffic Control.
Text Books:	
1.	IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017.
2.	The Definitive Guide to ARM Cortex-MR3 and M4 Processor, 3rd Edition, Joseph Yiu
Reference Books:	
1.	Internet of Things – A hands-on approach, Arshdeep Bahga, Vijay Madisetti, Universities Press, 2015.
2.	The Internet of Things – Key applications and Protocols, Olivier Hersent, David Boswarthick, Omar Elloumi and Wiley, 2012 (for Unit 2).
3.	“From Machine-to-Machine to the Internet of Things – Introduction to a New Age of Intelligence”, Jan Hoeller, Vlasios Tsiatsis, Catherine Mulligan, Stamatis, Karnouskos, Stefan Avesand. David Boyle and Elsevier, 2014.
4.	Architecting the Internet of Things, Dieter Uckelmann, Mark Harrison, Michahelles and Florian (Eds), Springer, 2011.
5.	Recipes to Begin, Expand, and Enhance Your Projects, 2nd Edition, Michael Margolis, Arduino Cookbook and O'Reilly Media, 2011.
6.	Cortex-A series Programmer's Guide for ARMv7-A by Arm. http://infocenter.arm.com/help/index.jsp?topic=/com.arm.doc.den0013d/index.html
7.	Internet of Things: Internet of Things: Architecture, Design Principles and Applications, Raj Kamal, McGraw Hill Higher Education Internet of Things,

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1206	PE	3	0	0	3	25	75	3 Hrs.
OPEN SOURCE PROGRAMMING								
(Program Elective-III)								
Course Objectives:								
1	To understand Open Source Programming concepts							
2	To build applications based on Open Source Software							
Course Outcomes: After completion of course, students will be able to:								
S.No	Outcome							Knowledge Level
1	Develop codes in open source web applications							K6
2	Understand the risks associated with the open source codes							K2
3	Write secure CGI scripts							K6
SYLLABUS								
UNIT-I (10 Hrs)	INTRODUCTION: Introduction to open source programming languages, advantages and drawbacks of open source programming, threats and vulnerabilities in open source languages, Operating System – Ubuntu Linux – Introduction to shell programming.							
UNIT-II (10 Hrs)	PHP: PHP Language Basics, Functions - calling a function, variable function, and anonymous function, Strings - cleaning, encoding and escaping, and comparing strings, Arrays – storing data in arrays, extracting multiple values, traversing, and sorting arrays, Objects – creation, introspection, and serialization, Web Techniques – processing forms and maintaining state.							
UNIT-III (10 Hrs)	WEB DATABASE APPLICATIONS: Three-tier architecture, Introduction to Object oriented programming with PHP 5, Database basics, MYSQL - querying web databases, writing to web databases, validation with Javascript, Form based authentication, protecting data on the web.							
UNIT-IV (10 Hrs)	PERL, TCL, AND PYTHON : Numbers and Strings, Control Statements, Lists and Arrays, Files, Pattern matching, Hashes, Functions. Introduction to TCL/TK, Introduction to Python.							
UNIT-V (10 Hrs)	SECURITY IN WEB APPLICATIONS : Recognizing web application security threats, Code Grinder, Building functional and secure web applications, Security problems with Javascript, vulnerable GCI scripts, Code Auditing and Reverse Engineering, types of security used in applications.							
Text Books:								
1.	Kevin Tatroe, Peter MacIntyre, Rasmus Lerdorf, “Programming PHP”, O’Reilly Media, 2012.							
2.	Michael Cross, “Developer’s Guide to Web Application Security”, Syngress Publishers, 2007.							

3.	Hugh E. Williams, David Lane, “Web Database applications with PHP and MYSQL”, Second Edition, O’Reilly Media, 2004.
Reference Books:	
1.	Tom Christiansen, Brian D Foy, Larry Wall, Jon Orwant, “Programming Perl”, Fourth Edition, O’Reilly Media, 2012.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1207	PE	3	0	0	3	25	75	3 Hrs.
PATTERN RECOGNITION								
(Program Elective-III)								
Course Objectives:								
1.	To implement pattern recognition and machine learning theories							
2.	To design and implement certain important pattern recognition techniques							
3.	To apply the pattern recognition theories to applications of interest							
4.	To implement the entropy minimization, clustering transformation and feature ordering							
S.No	Outcome							Knowledge Level
1	Design systems and algorithms for pattern recognition (signal classification), with focus on sequences of patterns that are analyzed using, e.g., hidden Markov models (HMM)							K6
2	Analyze classification problems probabilistically and estimate classifier performance							K4
3	Understand and analyze methods for automatic training of classification systems,							K2
4	Apply Maximum-likelihood parameter estimation in relatively complex probabilistic models, such as mixture density models and hidden Markov models							K3
5	Understand the principles of Bayesian parameter estimation and apply them in relatively simple probabilistic models							K2
SYLLABUS								
UNIT-I (10 Hrs)	Introduction - Basic concepts, Applications, Fundamental problems in pattern Recognition system design, Design concepts and methodologies, Examples of Automatic Pattern recognition systems, Simple pattern recognition model, Decision and Distance Functions - Linear and generalized decision functions, Pattern space and weight space, Geometrical properties, implementations of decision functions, Minimum-distance pattern classifications.							
UNIT-II (10 Hrs)	Probability-Probability of events, Random variables, Joint distributions and densities, Movements of random variables, Estimation of parameter from samples, Statistical Decision Making - Introduction, Baye's theorem, Multiple features, Conditionally independent features, Decision boundaries, Unequal cost of error, estimation of error rates, the leaving-one-out-techniques, characteristic curves, estimating the composition of populations. Baye's classifier for normal patterns.							
UNIT-III (10 Hrs)	Non Parametric Decision Making - Introduction, histogram, kernel and window estimation, nearest neighbor classification techniques. Adaptive decision boundaries, adaptive discriminate functions, Minimum squared error Discriminate functions, choosing a decision making techniques. Clustering and Partitioning - Hierarchical Clustering: Introduction, agglomerative clustering algorithm, the single-linkage, complete-linkage and average-linkage algorithm. Ward's method Partition clustering-Forg's algorithm, K-means's algorithm, Isodata algorithm.							

UNIT-IV (10 Hrs)	Pattern Preprocessing and Feature Selection: Introduction, distance measures, clustering transformation and feature ordering, clustering in feature selection through entropy minimization, features selection through orthogonal expansion, binary feature selection.
UNIT-V (10 Hrs)	Syntactic Pattern Recognition & Application Of Pattern Recognition: Introduction, concepts from formal language theory, formulation of syntactic pattern recognition problem, syntactic pattern description, recognition grammars, automata as pattern recognizers, Application of pattern recognition techniques in bio-metric, facial recognition, IRIS scan, Finger prints, etc.,
Text Books:	
1.	Gose. Johnsonbaugh. Jost. “ Pattern recognition and Image Analysis”, PHI. Tou. Rafael. Gonzalez. “Pattern Recognition Principle”, Pearson Education
Reference Books:	
1.	Richard duda, Hart., David Strok, “Pattern Classification”, John Wiley

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1208	PE	3	0	0	3	25	75	3 Hrs.
NATURAL LANGUAGE PROCESSING								
(Program Elective-IV)								
Course Objectives:								
1.	Make them understand the concepts of morphology, syntax, semantics and pragmatics of the language and that they are able to give the appropriate examples that will illustrate the above mentioned concepts.							
2.	Teach them to recognize the significance of pragmatics for natural language understanding.							
3.	Enable students to be capable to describe the application based on natural language processing and to show the points of syntactic, semantic and pragmatic processing.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Explain approaches to syntax and semantics in NLP.							K2
2	Demonstrate approaches to discourse, generation, dialogue and summarization within NLP.							K3
3	Explain current methods for statistical approaches to machine translation.							K2
4	Identify machine learning techniques used in NLP, including hidden Markov models and probabilistic							K2
5	Explain context-free grammars, clustering and unsupervised methods, log-linear and discriminative							K2
SYLLABUS								
UNIT-I (10Hrs)	Introduction: NLP tasks in syntax, semantics, and pragmatics. Applications such as information extraction, question answering, and machine translation. The problem of ambiguity. The role of machine learning. Brief history of the field.							
UNIT-II (10Hrs)	N-gram Language Models: The role of language models, Simple Ngram models. Estimating parameters and smoothing. Evaluating language models. Part of Speech Tagging and Sequence Labeling: Lexical syntax. Hidden Markov Models. Maximum Entropy Models. Conditional Random Fields.							
UNIT-III (10Hrs)	Syntactic parsing: Grammar formalisms and tree banks. Efficient parsing for context-free grammars (CFGs). Statistical parsing and probabilistic CFGs (PCFGs). Lexicalized PCFGs.							
UNIT-IV (10Hrs)	Semantic Analysis: Lexical semantics and word-sense disambiguation. Compositional semantics. Semantic Role Labeling and Semantic Parsing.							

UNIT-V (10Hrs)	Information Extraction (IE) and Machine Translation (MT): Named entity recognition and relation extraction. IE using sequence labeling. Basic issues in MT. Statistical translation, word alignment, phrase based translation, and synchronous grammars. Dialogues: Turns and utterances, grounding, dialogue acts and structures Natural Language Generation: Introduction to language generation, architecture, discourse planning (text schemata, rhetorical relations).
Text Books:	
1.	D. Jurafsky & J. H. Martin – “Speech and Language Processing – An introduction to Language processing, Computational Linguistics, and Speech Recognition”, Pearson Education.
Reference Books:	
1	Allen, James. 1995. – “Natural Language Understanding”. Benjamin/ Cummings, 2ed.
2	Bharathi, A., Vineet Chaitanya and Rajeev Sangal. 1995. Natural Language Processing- “A Pananian Perspective”. Prentice Hall India, Eastern Economy Edition.
3	Eugene Charniak: “Statistical Language Learning”, MIT Press, 1993.
4	Manning, Christopher and Heinrich Schutze. 1999. “Foundations of Statistical Natural Language Processing”. MIT Press.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1209	PE	3	0	0	3	25	75	3 Hrs.
FULL STACK TECHNOLOGIES								
(Program Elective-IV)								
Course Objectives:								
1.	Translate user requirements into the overall architecture and implementation of new systems and Manage Project and coordinate with the Client.							
2.	Write backend code in Python/Java, PHP languages and Writing optimized front end code HTML and JavaScript.							
3.	Understand, create and debug database related queries and Create test code to validate the applications against client requirement.							
4.	Monitor the performance of web applications & infrastructure and Troubleshooting web application with a fast and accurate a resolution.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Identify the Basic Concepts of Web & Markup Languages							K2
2	Develop web Applications using Scripting Languages & Frameworks							K6
3	Creating and running applications using PHP							K6
4	Creating Our First Controller Working with and Displaying in Angular Js and Nested Forms with ng-form							K6
5	Working with the Files in React JS and Constructing Elements with Data							K6
SYLLABUS								
UNIT-I (10Hrs)	HTML Web Essentials: Clients, Servers, and Communication. The Internet-Basic Internet Protocols -The World Wide Web-HTTP request message-response message-Web Clients Web Servers. Markup Languages: XHTML,an Introduction to HTML, History, Versions, Basic, XHTML Syntax and Semantics Some Fundamental HTML Elements-Relative URLs-Lists-tables-Frames-Forms-HTML 5.0.							
UNIT-II (10Hrs)	Cascading Style Sheets (CSS) Style Sheets: CSS-Introduction to Cascading Style Sheets-Features-Core Syntax-Style Sheets and HTML- Style Rule Cascading and Inheritance-Text Properties-Box Model Normal Flow Box Layout beyond the Normal Flow- CSS3.0, Boot strap basics, Boot strap CSS3, Introduction to Java Script, Jscript basics, JScripts objects, JSON, Don.							
UNIT-III (10Hrs)	PHP Introduction to PHP, Language Basics, Functions, Strings, Arrays. MYSQL Installation, Accessing MySQL Using PHP, Form Handling, Cookies, Sessions, and Authentication, Tables, Inserting Data into Tables, Selecting Data from a Table, Updating Table, Deleting data from Table, Webpage creation.							

UNIT-IV (10Hrs)	Angular Js Introducing AngularJS, Starting Out with AngularJS, Basic AngularJS, Directives and Controllers, AngularJS Modules, Creating First Controller, working with and Displaying, Arrays, more Directives, working with ng-repeat, Unit Testing in AngularJS, Forms, Inputs, and Services, Working with ng-model, Working with Forms, Leverage Data-Binding and Models, Form Validation and States, Error Handling with Forms, ngModelOptions, Nested Forms with ng-form, Other Form Controls.
UNIT-V (10Hrs)	React JS Introduction to React, Obstacles and Roadblocks, keeping Up with the Changes, Working with the Files, Pure React, Page Setup, The Virtual DOM, React Elements, ReactDOM, Children, Constructing Elements with Data, React Components, DOM Rendering, Factories
Text Books:	
1.	Jeffrey C. Jackson, "Web Technologies--A Computer Science Perspective", Pearson Education, 2006
2.	Robert. W. Sebesta, "Programming the World Wide Web", Fourth Edition, Pearson Education, 2007
3.	AngularJS: Up and Running Enhanced Productivity with Structured Web Apps By Brad Green, Shyam Seshadri Publisher: O'Reilly Media
4.	Learning React Functional Web Development with React and Redux By <u>Alex Banks</u> , <u>Eve Porcello</u>
Reference Books:	
1	Head First Java, 2nd Edition by Bert Bates, Kathy Sierra Publisher: O'Reilly Media, Inc

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1210	PE	3	0	0	3	25	75	3 Hrs.
PARALLEL ALGORITHMS								
(Program Elective-IV)								
Course Objectives:								
1.	The student will learn How to work on Parallel Programming Platforms, Principles of Parallel Algorithm Design, Parallelization aspects , Parallel sorting methods, Mapping and scheduling aspects of algorithms							
Course Outcomes By the end of this course student will be able to								
S.No	Outcome							Knowledge Level
1.	Understand fundamental concepts of parallelism- pipeline, Amdahl's law.							K2
2.	Know the physical limits of linear approach and solving problems in parallel.							K2
3.	How to design & analyze parallel algorithms and implement them with parallel processors.							K6
4.	Understand various approaches in parallel sorting and Searching.							K2
5.	Gain knowledge on various parallel processor architectures and know how to embed one Architecture into another.							K1
SYLLABUS								
UNIT-I (10 Hrs)	Introduction : Computational demand in various application areas, advent of parallel processing, terminology-pipelining, Data parallelism and control parallelism-Amdahl’s law.							
UNIT-II (10 Hrs)	Scheduling: Organizational features of Processor Arrays, Multi processors and multi-computers. Mapping and scheduling aspects of algorithms. Mapping into meshes and hyper cubes-Load balancing-List scheduling algorithm Coffman-graham scheduling algorithm for parallel processors.							
UNIT-III (10 Hrs)	Algorithms : Elementary Parallel algorithms on SIMD and MIMD machines, Analysis of these algorithms. Matrix Multiplication algorithms on SIMD and MIMD models. Fast Fourier Transform algorithms. Implementation on Hyper cube architectures. Solving linear file -system of equations, parallelizing aspects of sequential methods back substitution and Tri diagonal.							
UNIT-IV (10 Hrs)	Sorting: Parallel sorting methods, Odd-even transposition Sorting on processor arrays, Biotonic ,merge sort on shuffle - exchange ID , Array processor,2D-Mesh processor and Hypercube Processor Array. Parallel Quick- sort on Multi processors. Hyper Quick sort on hypercube multi computers. Parallel search operations. Ellis algorithm and Manber and ladner’s Algorithms for dictionary operations.							

UNIT-V (10 Hrs)	Searching Parallel algorithms for Graph searching, All Pairs shortest paths and inimum cost spanning tree. Parallelization aspects of combinatorial search algorithms with Focus on Branch and Bound Methods and Alpha-beta Search methods.
Text Books:	
1.	Parallel computing t h e o r y and practice, Michel J.Quinn
2.	Programming Parallel Algorithms, Guy E. Blelloch, Communications of the ACM
Reference Books:	
1.	Introduction to Parallel Computing, 2e, Ananth Grama, Anushul Guptha, George Karypis, vipin kumar, Addison- Wesley.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1211	PE	3	0	0	3	25	75	3 Hrs.
OBJECT ORIENTED SOFTWARE ENGINEERING								
(Program Elective-IV)								
Course Objectives:								
1.	Gain Knowledge in both the principles of software engineering as well as the practices of various object- oriented tools, processes, and products.							
2.	Design and construction of modular, reusable, extensible and portable software using object oriented programming languages.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Apply object-oriented programming principles to real-time problems..							K3
2	Analyze of a formally specified problem statement with respect to its accuracy and completeness, to effective testing of the software product.							K4
3	Examine the specialised knowledge, skill and judgement needed to develop complex software by formulating relevant responses at each stage of the software development life-cycle.							K5
4	Produce appropriate documentation accurately and to a professional standard.							K4
5	Apply skills relevant for academic progression and career development within the sector.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Software Engineering- What is Software Engineering, Software Engineering Concepts, Software Engineering Development Activities, Managing Software Development, Case Study, Modeling with UML- Introduction – Overview of UML – Modeling Concepts – Deeper View into UML.							
UNIT-II (10Hrs)	Project Organization and Communications- Introduction, An Overview of Projects, Project Organization Concepts, Project Communication Concepts, Organizational Activities, Analysis- Introduction, Overview of Analysis, Analysis Concepts, Analysis Activities, Managing Analysis, Case study..							
UNIT-III (10Hrs)	System Design- Overview of System Design, System Design Concepts, System Design activities, Managing System Design, Case study, Object Design- Overview of Object design, Reuse Concepts, Reuse Activities, Managing Reuse, Case study.							
UNIT-IV (10Hrs)	Mapping Models to Code- Overview of mapping, Mapping concepts, Mapping Activities, Mapping Implementation, Case study, Configuration Management and Project Management- Configuration Management Overview, Concepts, Activities and Managing Configuration Management, Overview of Project management, Project Management Concepts, Project Management Activities.							

UNIT-V (10Hrs)	Software Life Cycle- Introduction, IEEE 1074, Characterizing the Maturity to Software Life Cycle Models, Life cycle Models, Methodologies- Introduction, Project Environment, Methodology Issues, A Spectrum of Methodologies, Case studies.
Text Books:	
1.	Object-Oriented Software Engineering: Practical software development using UML, Patterns and java, Second Edition, Bernd Bruegge and Allen Dutoit, Pearson Education, 2004. ISBN-10: 0130471100.
Reference Books:	
1	Object-Oriented Software Engineering: Conquering Complex and Changing Systems, Bernd Bruegge and Allen H. Dutoit, Pearson Education, 2002. ISBN 0-13-489725-0.
2	Object-oriented Software Engineering: The Professional Developer's Guide, Addison-Wesley, George Wilkie, 1993. ISBN-10: 0201627671.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1212	PE	3	0	0	3	25	75	3 Hrs.
DISTRIBUTED DATABASES								
(Program Elective-IV)								
Course Objectives:								
1.	The aim of this module is to build on the previous background of database systems by Deepening the understanding of the theoretical and practical aspects of the database technologies showing the need for distributed database technology to tackle deficiencies of the centralized database systems and finally introducing the concepts and techniques of distributed database including principles, architectures, design, implementation and major domain of application.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Identify the introductory distributed database concepts and its structures.							K1
2	Describe terms related to distributed object database design and management.							K1
3	Produce the transaction management and query processing techniques in DDBMS.							K4
4	Relate the importance and application of emerging database technology.							K4
SYLLABUS								
UNIT-I (10Hrs)	Introductory concepts and design of (DDBMS) Data Fragmentation; Replication; and allocation techniques for DDBMS; Methods for designing and implementing DDBMS, designing a distributed relational database; Architectures for DDBMS: cluster federated, parallel databases and client server architecture.							
UNIT-II (10Hrs)	Query processing & Transaction Management Overview Of Query Processing: Query processing problem; Objectives of Query Processing; Complexity of Relational Algebra operations; characterization of Query processors; Layers of Query Processing Introduction To Transaction Management: Definition of Transaction, Properties of Transaction, types of transaction ; Distributed Concurrency Control: Serializability theory; Taxonomy of concurrency control mechanisms; locking based concurrency control algorithms.							
UNIT-III (10Hrs)	Distributed Object Database Management systems Fundamental Object concepts and Object models; Object distribution design; Architectural issues; Object management; Distributed object storage; Object query processing.							
UNIT-IV (10Hrs)	Current trends & developments related to Distributed database applications technologies Distributed Object/component-based DBMS; Database Interoperability including CORBA; DCOM and Java RMI; Distributed document-based systems; XML and Workflow management							

UNIT-V (10Hrs)	Emerging related database technologies Parallel Database; Mobile database; Multimedia Database; Spatial Database and Web Databases.
Text Books:	
1.	Distributed Databases - Principles and Systems; Stefano Ceri; Guiseppe Pelagatti; Tata McGraw Hill; 1985.
2.	Fundamental of Database Systems; Elmasri & Navathe; Pearson Education; Asia
3.	Database System Concepts; Korth & Sudarshan; TMH
4.	Principles of Distributed Database Systems; M. Tamer Özsu; and Patrick Valduriez Prentice Hall
Reference Books:	
1.	Data Base Management System; Leon & Leon; Vikas Publications
2.	Introduction to Database Systems; Bipin C Desai; Galgotia

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1213	PC	0	0	4	2	25	75	3 Hrs.
ADVANCE ALGORITHMS LAB								
Course Objectives:								
To Understand								
1.	Knowing about oops concepts for a specific problem							
2.	Various advanced data structures concepts like arrays, stacks, queues, linked lists, graphs and trees.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Identify classes, objects, members of a class and relationships among them needed for a specific problem.							K3
2.	Examine algorithms performance using Prior analysis and asymptotic notations.							K3
3.	Organize and apply to solve the complex problems using advanced data structures (like arrays, stacks, queues, linked lists, graphs and trees.)							K3
4.	Apply and analyze functions of Dictionary							K3
LIST OF EXPERIMENTS								
1.	Write a program to implement topological sorting.							
2.	Write a program to find inverse of a matrix.							
3.	Write a program to implement Floyd-Warshalls’ algorithm using dynamic programming							
4.	Write a program to find longest common subsequence of two strings							
5.	Write a program to implement BFS for a Graph							
6.	Write a program to use DFS to find strongly connected components of a digraph.							
7.	Write a program to implement matrix chain multiplication.							
8.	Write a program to implement randomized Quick Sort.							
9.	Write a program to solve Travelling Salesperson Problem using twice around the tree algorithm (approximation algorithm).							
10.	Write a program to implement FFT.							
11.	Write a program to implement Krushkal’s algorithm to generate a min-cost spanning tree.							
12.	Write a program to implement Prim’s algorithm to generate a min-cost spanning tree.							
13.	Write a program to implement Extended Euclid algorithm to find gcd of two numbers.							
Reference Books:								
1.	Algorithm Design, J. Kleinberg and E.Tardos, Pearson(Addison Wesley)							
2.	Computer Algorithms, Introduction to design and analysis, 3rd Edition , Sara Baase, Allen, Van, Gelder, Pearson Education							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1214	PC	0	0	4	2	25	75	3 Hrs.
MACHINE LEARNING LAB								
Course Objectives:								
1.	To learn and understand different Data sets in implementing the machine learning algorithms.							
2.	Implement the machine learning concepts and algorithms in any suitable language of choice.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Implement procedures for the machine learning algorithms							K3
2.	Apply appropriate data sets to the Machine Learning algorithms							K3
3.	Identify and apply Machine Learning algorithms to solve real world problems							K1
4.	Design Python programs for various Learning algorithms							K6
LIST OF EXPERIMENTS								
Experiments: Note: Implement using R and Python								
1.	Write a program to apply the following machine learning methods on any chosen dataset: a) Linear Regression b) Logistic Regression.							
2.	Write a program to implement Support Vector Machines.							
3.	Perform Exploratory Data Analysis for Classification using Pandas and Matplotlib.							
4.	Write a program to implement to analyze Bias, Variance, and Cross Validation.							
5.	Write a program to simulate a perception network for pattern classification and function approximation.							
6.	Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.							
7	Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.							
8	Write a program to implement the naïve Bayesian classifier for Iris data set. Compute the accuracy of the classifier, considering few test data sets.							
9	Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task. Built-in Java classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set.							
10	Apply EM algorithm to cluster a Heart Disease Data Set. Use the same data set for clustering using k-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. You can add Java/Python ML library classes/API in the program.							
11	Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions.							
12	Implementing data visualization using R Find the data distributions using box and scatter plot. Find the outliers using plot. Plot the histogram, bar chart and pie chart on sample data.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST1215	MP	0	0	4	2	100	--	3 Hrs.
MINI PROJECT WITH SEMINAR								
For Mini Project with Seminar, a student under the supervision of a faculty member, shall collect the literature on a topic and critically review the literature and submit it to the department in a report form and shall make an oral presentation before the Project Review Committee consisting of Head of the Department, supervisor/mentor and two other senior faculty members of the department. For Mini Project with Seminar, there will be only internal evaluation of 100 marks. A candidate has to secure a minimum of 50% of marks to be declared successful.								

Code	Category	L	T	P	C	I.M	E.M	Exam
#AC-2	AC	2	0	0	0	0	0	--
AUDIT COURSE-2								
List of Audit Courses and their Syllabi are mentioned in the First Semester Syllabus. The students can opt any one course for AC 2 from the list mentioned in first semester by not opting the course which is already taken for AC 1								



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, All UG Programmes are accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION (Regulation R19) **M.TECH (COMPUTER SCIENCE & TECHNOLOGY)** **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING** (With effect from 2019-2020 Admitted Batch onwards)

III-SEMESTER

Subject Code.	Name of the Subject	Category	C	L	T	P	Internal Marks	External Marks	Total Marks
#PE-V/ MOOCS	Program Elective-V	PE	3	3	0	0	25	75	100
#OE-I/ MOOCS	Open Elective-I	OE	3	3	0	0	25	75	100
M19CST 2105	Dissertation-I / Industrial Project	PR	10	0	0	20	50	50	100
TOTAL			16	6	0	20	100	200	300

	Code	Course
#PE-V	M19CST2101	Deep Learning
	M19CST2102	Ethical Hacking
	M19CST2103 (MOOCS-I)	Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course
#OE-I	OE	Students have to choose one open elective course offered by departments other than the CSE & IT departments. List of open Electives offered by other departments are enclosed.
	M19CST2104 (MOOCS-II)	Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course

OPEN ELECTIVES OFFERED TO OTHER DEPARTMENTS

Code	Course
M19 CST 2106	Python Programming
M19 CST 2107	Artificial Intelligence
M19 CST 2108	Advanced Data structures

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST2101	PE	3	0	0	3	25	75	3 Hrs.
DEEP LEARNING								
(PROGRAM ELECTIVE-V)								
Course Objectives:								
1.	Learn deep learning methods for working with sequential data							
2.	Learn deep recurrent and memory networks							
3.	Apply such deep learning mechanisms to various learning problems							
4.	Learn deep Turing machines, the open issues in deep learning, and have a grasp of the current research directions.							
Course Outcomes								
After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
1.	Demonstrate the basic concepts fundamental learning techniques and layers.							K2
2.	Discuss the Neural Network training, various random models.							K3
3.	Explain different types of deep learning network models.							K2
4.	Classify the Probabilistic Neural Networks.							K3
5.	Implement tools on Deep Learning techniques.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Various paradigms of learning problems, Perspectives and Issues in deep learning framework, review of fundamental learning techniques. Feed forward neural network: Artificial Neural Network, activation function, multi-layer neural network							
UNIT-II (10 Hrs)	Training Neural Network: Risk minimization, loss function, back propagation, regularization, model selection, and optimization. Deep Neural Networks: Difficulty of training deep neural networks, Greedy layer wise training.							
UNIT-III (10 Hrs)	Deep Learning: Deep Feed Forward network, regularizations, training deep models, dropouts, Convolution Neural Network, Recurrent Neural Network, and Deep Belief Network.							
UNIT-IV (10 Hrs)	Probabilistic Neural Network: Hopfield Net, Boltzmann machine, RBMs, Sigmoid net, Auto encoders.							
UNIT-V (10 Hrs)	Applications: Object recognition, sparse coding, computer vision, natural language processing. Introduction to Deep Learning Tools: Tensor Flow, Caffe, Theano, Torch.							
Text Books:								
1.	Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press, 2016							
2.	Bishop, C. ,M., Pattern Recognition and Machine Learning, Springer, 2006.							

Reference Books:	
1.	Yegnanarayana, B., Artificial Neural Networks PHI Learning Pvt. Ltd, 2009.
2.	Golub, G.,H., and Van Loan,C.,F., Matrix Computations, JHU Press,2013.
3.	Satish Kumar, Neural Networks: A Classroom Approach, Tata McGraw-Hill Education, 2004
4.	Neural Networks: A Systematic Introduction, Raúl Rojas, 1996
5.	Pattern Recognition and Machine Learning, Christopher Bishop, 2007

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST2102	PE	3	0	0	3	25	75	3 Hrs.
ETHICAL HACKING								
(Program Elective-V)								
Course Objectives:								
1.	To gain knowledge about Ethical hacking and penetration testing.							
2.	To learn about various types of attacks, attackers and security threats and vulnerabilities present in the computer system.							
3.	To examine how social engineering can be done by attacker to gain access of useful & sensitive information about the confidential data.							
4.	To learn about cryptography, and basics of web application attacks.							
5.	To gain knowledge of the tools , techniques and ethical issues likely to face the domain of ethical hacking and ethical responsibilities.							
Course Outcomes								
After completion of course, students would be able to:								
S.No	Outcome							Knowledge Level
CO1	Remember various hacking methods, system security vulnerability testing.							K1
CO2	Apply system vulnerability attacks and demonstrate a security assessment report							K3
CO3	Understand various issues related to hacking							K2
SYLLABUS								
UNIT-I (10 Hrs)	Hacking Windows: BIOS Passwords, Windows Login Passwords, Changing Windows Visuals, Cleaning Your Tracks, Internet Explorer Users, Cookies, URL Address Bar, Netscape Communicator, Cookies URL History, The Registry, Baby Sitter Programs.							
UNIT-II (10 Hrs)	Advanced Windows Hacking: Editing your Operating Systems by editing Explorer.exe, The Registry, The Registry Editor, Description of .reg file, Command Line Registry Arguments, Other System Files, Some Windows & DOS Tricks, Customize DOS, Clearing the CMOS without opening your PC, The Untold Windows Tips and Tricks Manual, Exiting Windows the Cool and Quick Way, Ban Shutdowns: A Trick to Play, Disabling Display of Drives in My Computer, Take Over the Screen Saver, Pop a Banner each time Windows Boots, Change the Default Locations, Secure your Desktop Icons and Settings.							
UNIT-III (10 Hrs)	Getting Past the Password: Passwords: An Introduction, Password Cracking, Cracking the Windows Login Password, The Glide Code, Windows Screen Saver Password, XOR, Internet Connection Password, Sam Attacks, Cracking Unix Password Files, HTTP Basic Authentication, BIOS Passwords, Cracking Other Passwords.							
UNIT-IV (10 Hrs)	The Perl Manual: Perl: The Basics, Scalars, Interacting with User by getting Input, Chomp() and Chop(), Operators, Binary Arithmetic Operators, The Exponentiation Operator(**), The Unary Arithmetic Operators, Other General Operators, Conditional Statements, Assignment Operators. The : Operator, Loops, The While Loop, The For Loop, Arrays, THE FOR EACH LOOP: Moving through an Array, Functions Associated with Arrays, Push() and							

	Pop(), Unshift() and Shift(), Splice(), Default Variables, \$_, @ARGV, Input Output, Opening Files for Reading, Another Special Variables.
UNIT-V (10 Hrs)	Virus Working, Boot Sector Viruses (MBR or Master Boot Record), File or Program Viruses, Multipartite Viruses, Stealth Viruses, Polymorphic Viruses, Macro Viruses, Blocking Direct Disk Access, Recognizing Master Boot Record (MBR) Modifications, Identifying Unknown Device Drivers, making own Virus, Macro Viruses, Using Assembly to Create your own Virus, Modifying a Virus so Scan won't Catch it, Creating New Virus Strains, Simple Encryption Methods.
Text Books:	
1.	Patrick Engbreston: "The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made Easy", 1st Edition, Syngress publication, 2011.
2.	Ankit Fadia : "Unofficial Guide to Ethical Hacking", 3rd Edition , McMillan India Ltd, 2006.
Reference Books:	
1.	Simpson/backman/corley, "HandsOn Ethical Hacking & Network Defense International", 2nd Edition, Cengageint, 2011.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CST 2103	PE	3	0	--	3	25	75	3 Hrs.
MOOCS-I								
Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course								

Code	Category	L	T	P	C	I.M	E.M	Exam
	OE	--	--	--	3	25	75	3 Hrs.
OPEN ELECTIVE								
Students have to choose one open elective course offered by departments other than the CSE & IT departments. List of open Electives offered by other departments are given below.								

Offered from	Course Code	Course Name	Offered to
CIVIL ENGINEERING	M19 ST 2107	Construction Management	CST, CS, PSA,IT & CAD/CAM
	M19 ST 2108	Green Technology	
	M19 ST 2109	Analysis of Offshore Structures	
ELECTRONICS & COMMUNICATION ENGINEERING	M19 CS 2107	Signals and systems	ST, CST, PSA, IT & CAD/CAM
	M19 CS 2108	Principles of Communication	
	M19 CS 2109	Image and video Processing	
ELECTRICAL & ELECTRONICS ENGINEERING	M19PS2107	Electric And Hybrid Vehicles	ST, CST, CS, IT & CAD/CAM
	M19PS2108	Energy From Waste	
	M19PS2109	Energy Management and Auditing	
MECHNAICAL ENGINEERING	M19CAD 2107	Operations Research	ST, CST, CS, PSA & IT
	M19CAD 2108	Nano Technology	
	M19CAD 2109	Product Design & Manufacturing	
SCIENCE & HUMANITIES	M19BS2101	Management and Organisational Behaviour	ST, CST, CS, PSA, IT & CAD/CAM

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST2104	PE	--	--	--	3	25	75	3 Hrs.
MOOCS-II								
Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course								

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST2105	PR	0	0	20	10	50	50	3 Hrs.
DISSERTATION-I/INDUSTRIAL PROJECT								
The Student has to register for Dissertation-I / Industrial project in III semester. Student has to submit, in consultation with his project supervisor, the title, objective and plan of action of his project work for approval. The student can initiate the Project work, only after obtaining the approval from the Project Review Committee (PRC). Continuous assessment of Dissertation-I during the III-Semester will be monitored by the PRC. Dissertation-I/ Industrial Project is evaluated for 50 internal marks and 50 external marks. Internal marks 50 awarded by Project Guide and PRC jointly based on continuous assessment consisting of two seminars based on Dissertation work-I. External marks 50 awarded by External Examiner, Supervisor and Head of the Department jointly based on a review and Viva voce on Dissertation work-I.								

SCHEME OF INSTRUCTION & EXAMINATION
 (Regulation R19)
M.TECH (COMPUTER SCIENCE & TECHNOLOGY)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
 (With effect from **2019-2020**Admitted Batch onwards)

IV-SEMESTER

Subject Code.	Name of the Subject	Category.	C	L	T	P	Internal Marks	External Marks
M19CST2201	Dissertation-II / Industrial Project	PR	16	0	0	32	--	100

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CST2201	PR	0	0	32	16	--	100	3 Hrs.
DISSERTATION-II/INDUSTRIAL PROJECT								
The student has to continue his/her work from Dissertation-I / Industrial project to complete Dissertation-II in IV semester. Continuous assessment of Dissertation-II during IV-Semester will be monitored by the PRC. Dissertation-II is evaluated for 100 external marks based on Review and Viva Voce. Review and Viva-Voce examination shall be conducted by a board consisting of the Supervisor, Head of the Department and the examiner who adjudicated the Thesis. The Board shall jointly report the candidate's work for 100 marks. If the report of the Viva-Voce is unsatisfactory (ie, < 50 marks), the candidate shall retake the Viva-Voce examination only after three months. If he fails to get a satisfactory report at the second Viva-Voce examination, the candidate has to reregister for the project and complete the project within the stipulated time after taking the approval from the College.								